

Formal Reductions and Singular Limits in Polylogarithm Arithmetic

Cyclotomic coefficient ranks, rational Herglotz values,
Lerch zero geometry, and complete angular asymptotics

A research continuation prepared for Vladimir Reshetnikov

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Abstract

We continue the ProveIt polylogarithm manuscript at a fixed repository snapshot, distinguishing newly proved results from recently completed incoming research. We prove its odd-weight Gaussian coefficient-rank conjecture, determine the even-weight ranks, and construct a canonical same-point splitting at every cyclotomic level. For the companion Herglotz integral $J(p/q)$, we give a complete rational-dilogarithm and logarithmic classification in the manuscript's rationalized five-term calculus, together with explicit positive-real evaluations. For differentiated Lerch logarithmic sums, we classify every small-parameter zero pattern, prove the optimal uniform threshold $K_2 = 1$, exhibit a nonmonotone zero branch, and compute the first singular Abel coefficient and the exact endpoint regularity of simple zero branches. Finally, we construct the angular-zero expansion to every exponential order, uniformly in the second index, and prove that its untruncated series diverges at every fixed first index. Exact coefficient generators, rational sign certificates, independent numerical diagnostics, a narrow editorial patch, and a proposed research program accompany the proofs. Formal relation ranks and five-term obstructions are kept distinct from unproved independence statements about numerical periods.

Source baseline: ProveIt commit `a2a4cf58c49c745058c40e4a6748d472a3420f18`.

AI-assisted research preparation; proofs and verification scope are documented explicitly. This article is an additive research contribution for review and integration.

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1 Scope, antecedents, and principal results

The starting point is the manuscript in `Analysis/Polylogarithms/docs/manuscript` of the ProveIt repository, at commit `a2a4cf58c49c745058c40e4a6748d472a3420f18` [1]. The associated incoming reports were inspected at the same commit. This matters mathematically: several questions that still appear open in older chapter text have already been answered in those reports. In particular, the Gaussian weight-five and weight-six reductions, the residual level-three and level-four polynomial spaces, and the Möbius-duality certificate for S_4 are antecedents, not results newly established here [2, 3].

The continuation develops four connected directions. Each replaces a finite observation or a qualitative boundary statement by an exact structural result. Polynomial symmetry determines an infinite family of matrix ranks. Exterior symbols determine the formal reduction type of every positive rational Herglotz argument. A singular perturbation analysis explains why real zeros disappear near the elementary Lerch endpoint and why the surviving branches need not be monotone. Finite Lagrange inversion gives the entire exponential scale of an angular zero, while a separate argument shows that the resulting infinite expansion is necessarily divergent.

1.1 Four different meanings of an exact assertion

We use the following distinctions throughout.

1. An *analytic identity* is an equality of specified functions or numbers, with domains, branches, constants, and limiting operations fixed. The explicit formulas for $J(3/5)$ and $J(n/(n+1))$ have this meaning.
2. A *coefficient-rank theorem* concerns a fully specified rational matrix generated by shuffle and stuffle rows. Its kernel is not a space of numerical periods. A complete answer for that matrix does not exclude identities arising from additional relations.
3. A *formal dilogarithm classification* concerns the rationalized pre-Bloch group and its boundary map. A nonzero obstruction excludes a reduction in that calculus. It does not prove that a numerical coincidence between periods is impossible.
4. An *exact certificate* is a finite rational or integer calculation; when it certifies an infinite analytic expression, it is accompanied by a stated error theorem. Floating-point root plots and high-precision identity checks are supplementary diagnostics; they are never used as substitutes for the analytic proofs.

These distinctions allow strong completeness statements without silently assuming transcendence or period conjectures.

1.2 The rank conjecture and its universal extension

The source's Conjecture `signed:conj:rank` uses $6w - 7$ imaginary depth-two coordinates at fourth roots of unity. It removes the divergent coordinate and retains a single regularized difference at the boundary. Theorem 2.1 proves both proposed odd-weight ranks and also gives the even-weight answer. More significantly, Theorem 2.4 describes the entire kernel through two freely chosen homogeneous polynomials. The previously unresolved issue is the canonical lift of the same-point polynomial through the precise boundary convention; a rank fit in low weights cannot supply this step.

Theorem 3.1 shows that the lift is independent of the cyclotomic level. It splits off one explicit antireflection sector for each conjugate pair of nonreal roots. At level three, combining this new lift with the earlier residual invariant ring gives the full ranks. The results provide exact separating functionals for proposed rows, as well as completeness of the same-point rows

in the stated system. They should be compared with the broader standard relations studied by Zhao [4] and with parity reductions [5, 6]; none of those broader theories is being identified with this restricted single-product matrix.

1.3 All rational arguments of the Herglotz companion

Let

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt, \quad x > 0.$$

For reduced p/q with both entries odd, Theorem 4.5 proves that a formal rational-dilogarithm reduction exists precisely when $p, q \in \{1, 3, 5\}$. A formal logarithmic reduction in this parity class occurs only at $p = q = 1$. If e is the even entry and o the odd entry, the two reduction types coincide and are characterized by

$$e^2 \equiv \pm 1 \pmod{o}, \quad o^2 \equiv 1 \pmod{2e}.$$

The doubled modulus is essential. The proof adds a proper-real-subfield faithfulness theorem for an individual cyclotomic symbol and an exact dyadic norm identity to the source's all-conductor Gram theorem.

The analytic evaluation section fixes the logarithmic and π^2 terms suppressed by the formal notation. It derives an explicit log-sine formula for every consecutive rational argument and obtains

$$\begin{aligned} J(3/5) = & \operatorname{Li}_2(1/3) - \frac{1}{2} \operatorname{Li}_2(1/5) - \frac{7\pi^2}{180} + \frac{1}{2} \log^2 2 + \frac{1}{2} \log^2 3 \\ & - \frac{1}{4} \log^2 5 + \log^2 \left(\frac{1+\sqrt{5}}{2} \right). \end{aligned}$$

The functional equations of Radchenko–Zagier and Choie–Kumar are credited explicitly [7, 8]. The finite difference used here is a reformulation of known functional equations, not a claimed new general functional equation. The classifications and explicit specializations are the contributions of this continuation.

1.4 Zero geometry at both Lerch endpoints

For $n, k \geq 1$, the normalized differentiated logarithmic Lerch sum is

$$G_{n,k}(a, \rho) = \sum_{m \geq 0} \rho^m \frac{P_{n,k}(-\log(a+m))}{(a+m)^{k+1}}, \quad a > 0, \quad 0 \leq \rho \leq 1.$$

The polynomial $P_{n,k}$ is explicit. At $\rho = 0$ it may have a multiple zero at $a = 1$. Theorem 8.2 resolves that zero into convergent Puiseux branches and gives the exact number of positive zeros for every fixed n, k and all sufficiently small positive ρ . A common compact zero region proves that the local analysis captures every positive zero.

For $n = 2$ we go further: there are exactly two simple positive zeros for every $k \geq 1$ and every $\rho \in [0, 1]$. Thus the source's uniform saturation threshold has the sharp value $K_2 = 1$. At $k = 1$ the smaller zero first moves below 1 and ends above 1, so it is nonmonotone. The larger branch is strictly decreasing; the smaller branch crosses 1 exactly once in the interior and increases thereafter. At the opposite endpoint, a universal term $\tau^k \log^{n+1}(1/\tau)$, with $\tau = -\log \rho$, makes every simple endpoint zero exactly C^{k-1} and not C^k . For the two $n = 2, k = 1$ branches the slopes tend respectively to $+\infty$ and $-\infty$.

1.5 Every angular scale, and why the series diverges

Let $\theta_{a,b}$ be the source’s unique zero of $\operatorname{Im} \operatorname{Li}_{a,b}(e^{i\theta}, 1)$ in the upper semicircle. We construct the expansion of $\delta_{a,b} = \pi/2 - \theta_{a,b}$ to any prescribed exponential accuracy, uniformly for all real $b > 0$. Its scales are products of the rational numbers $2/n$, $n \geq 3$, and its coefficients are exact polynomials in generalized harmonic numbers. Every cutoff requires only finitely many coefficients. Ten explicit terms and a sharp signed remainder are tabulated.

The final asymptotic theorem also clarifies what “complete” means. At each fixed a, b , a prime-indexed subsequence of the grouped terms does not even tend to zero. Consequently the infinite series cannot be summed by ordinary convergence, although every fixed exponential cutoff is rigorously asymptotic as $a \rightarrow \infty$. This is a concrete example in which exact, arbitrarily long expansions coexist with unavoidable divergence.

1.6 Dependencies and reproducibility

The matrix proofs and angular coefficient construction are algebraic and analytic arguments valid in all weights. The Herglotz classification uses the source’s proved Gram theorem and its explicitly stated rationalized Bloch-group facts. The zero results use the source’s global variation bound and eleven replayed rational endpoint signs; the essential background and the variation argument are recalled here. Only the nonvanishing of a polynomial at $-\log 2$ uses classical Hermite–Lindemann transcendence.

The source manifest, scripts, output receipts, and a claim ledger are included with the article. New proofs received independent internal mathematical cross-checks. No claim of formal proof-assistant verification or external peer review is made. The remaining questions are listed with concrete next steps in Section 14.

2 The full Gaussian single-product rank conjecture

The manuscript’s Conjecture `signed:conj:rank` concerns a precise coefficient matrix, rather than the dimension of a space of actual polylogarithmic numbers. We prove both proposed rank formulas in every odd weight. The proof also gives the full even-weight answer, an explicit basis of the nullspace, and a complete finite test for whether an arbitrary coefficient vector follows from the stated rows.

The additional boundary issue is essential. A preceding incoming report, `polylogarithms_research_2026-10-10.zip`, proves the dimension after setting the Gaussian family to zero and after retaining regularized rows in a different presentation. Our result keeps that family free, removes exactly the manuscript’s divergent coordinate, and retains only the shuffle-minus-shuffle difference when a single factor diverges. The new step is a canonical lift of every allowed Gaussian nullvector through that precise boundary presentation.

2.1 The matrix and its scope

Fix an integer $w \geq 2$ and write $n = w - 2$. For $a + b = w$, let

$$D_{a,b}(x, y) = \sum_{k \geq \ell > 0} \frac{x^k y^\ell}{k^a \ell^b}, \quad X_a^{r,s} = \operatorname{Im} D_{a,w-a}(i^r, i^s), \quad r, s \in \mathbb{Z}/4\mathbb{Z}.$$

For the coefficient computation, the $X_a^{r,s}$ are formal rational coordinates. Impose conjugation $X_a^{-r,-s} = -X_a^{r,s}$ and put $X_a^{r,s} = 0$ if both colors are even. A set of representatives of the remaining color pairs is

$$(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1). \tag{2.1}$$

Remove $X_1^{0,1}$, the single divergent imaginary coordinate in these representatives. Thus the ambient coefficient space has dimension $N_w = 6(w-1) - 1 = 6w - 7$.

For $p+q=w$ and $r, s \in \mathbb{Z}/4\mathbb{Z}$, define the stuffle and shuffle coefficient rows

$$S_{p,q}^{r,s} = X_p^{r,s} + X_q^{s,r}, \quad (2.2)$$

$$H_{p,q}^{r,s} = \sum_{j=0}^{p-1} \binom{q+j-1}{j} X_{q+j}^{s,r-s} + \sum_{j=0}^{q-1} \binom{p+j-1}{j} X_{p+j}^{r,s-r}. \quad (2.3)$$

If both single factors $\text{Li}_p(i^r)$ and $\text{Li}_q(i^s)$ are finite, retain the two rows separately. If exactly one factor is $\text{Li}_1(1)$, retain only $S_{p,q}^{r,s} - H_{p,q}^{r,s}$, in which the removed coordinate cancels. The two-divergent-factor case occurs only at $w=2$ with two real colors; its imaginary coefficient row is zero. Discard zero rows; retaining repeated rows does not change the problem. Call the resulting rational matrix A_w . Let A_w^{-g} be obtained by deleting the $w-1$ columns $X_a^{1,0}$.

The depth-one right sides are suppressed because the present question concerns the coefficient row space. Analytically, these rows come from convergent shuffle and stuffle identities, and from subtracting the one-divergence identities before taking a regularized limit. Nothing here assigns a value to $\zeta(1)$. This is a restricted linear system of products of two depth-one values. In particular, it omits distribution, higher-depth products and elimination, and additional argument transformations. It is narrower than the families of standard relations studied by Zhao; see J. Zhao, *Standard Relations of Multiple Polylogarithm Values at Roots of Unity*, Documenta Mathematica **15** (2010), 1–34, [arXiv:0707.1459](#).

Theorem 2.1 (Exact ranks in every weight). *The matrix family just defined satisfies*

$$\text{rank}_{\mathbb{Q}} A_w = \begin{cases} 6w-7, & w \text{ even}, \\ 5w-6, & w \text{ odd}, \end{cases} \quad (2.4)$$

$$\text{rank}_{\mathbb{Q}} A_w^{-g} = \begin{cases} 5w-6, & w \text{ even}, \\ (9w-11)/2, & w \text{ odd}. \end{cases} \quad (2.5)$$

In particular, both equalities in the manuscript's odd-weight rank conjecture hold for every odd $w \geq 3$.

The proof is an explicit description of the entire kernel. It does not extrapolate from finite rank calculations.

2.2 Encoding the rows by homogeneous polynomials

For a scalar assignment to the formal coordinates, set

$$\mathcal{F}^{r,s}(X, Y) = \sum_{a=1}^{w-1} X_a^{r,s} X^{a-1} Y^{w-a-1}.$$

These homogeneous polynomials have degree n . Write

$$U = \mathcal{F}^{0,1}, \quad G = \mathcal{F}^{1,0}, \quad A = \mathcal{F}^{1,1}, \quad K = \mathcal{F}^{1,2}, \quad M = \mathcal{F}^{1,3}, \quad V = \mathcal{F}^{2,1}.$$

The letter A in these formulas denotes a polynomial; the matrix is always written A_w . The removed coordinate is encoded by

$$[Y^n]U(X, Y) = 0. \quad (2.6)$$

For a polynomial P , let $P^\sigma(X, Y) = P(Y, X)$.

Coefficient comparison turns a complete set of finite stuffle rows into $\mathcal{F}^{r,s}(X, Y) + \mathcal{F}^{s,r}(Y, X) = 0$. Similarly, the shuffle rows are the coefficients of

$$\mathcal{F}^{s,r-s}(X + Y, X) + \mathcal{F}^{r,s-r}(X + Y, Y).$$

For example, the coefficient of $X^{p-1}Y^{q-1}$ in the second summand is precisely the second binomial sum in (2.3). This explains the substitutions without invoking any rank or transcendence theorem.

Lemma 2.2 (The boundary defect). *Put $c = [X^n]G(X, Y)$. The finite rows with product colors $(0, 1)$, together with their one-divergence difference row, are exactly*

$$U(X, Y) + G(Y, X) = cY^n, \quad (2.7)$$

$$M(X + Y, X) + U(X + Y, Y) = cY^n. \quad (2.8)$$

Consequently they determine

$$U(X, Y) = -G(Y, X) + cY^n, \quad M(X, Y) = G(X - Y, X). \quad (2.9)$$

Proof. For colors $(0, 1)$, the factor $\text{Li}_p(1)$ is finite exactly when $p \geq 2$. Therefore the coefficients of $X^{p-1}Y^{q-1}$ in the stuffle polynomial $U(X, Y) + G(Y, X)$ vanish for all $p \geq 2$. The only possible defect is a multiple of Y^n . By (2.6), its coefficient is c . The same finite-product argument shows that $M(X + Y, X) + U(X + Y, Y)$ has only a Y^n defect, say dY^n . When $p = 1$, the retained regularized row is the difference of these two coefficient rows. It says exactly $c - d = 0$. This proves (2.8). Substituting the first identity into the second cancels cY^n and gives $M(X + Y, X) = G(Y, X + Y)$. The invertible substitution $(X + Y, X) \mapsto (X, Y)$ yields (2.9). The argument also covers $w = 2$: there are no $p \geq 2$ coefficients, and the difference row still equates the two constant defects. $\square$

This lemma is the point at which the precise regularization convention matters. Setting the two boundary rows separately to zero would impose an extra condition on c . Deleting the boundary row altogether would lose a condition. The manuscript's difference convention produces the unique correction cY^n .

Lemma 2.3 (Exhaustive kernel conditions). *An assignment annihilates every row of A_w if and only if its six polynomials satisfy (2.9) and*

$$A(Y, X) = -A(X, Y), \quad V(X, Y) = -K(Y, X), \quad (2.10)$$

$$M(Y, X) = M(X, Y), \quad G(X, Y) = -G(X, X - Y), \quad (2.11)$$

$$K(X, Y) = -A(Y, Y - X), \quad K(X, Y) = K(X, X - Y). \quad (2.12)$$

Proof. The product-color classes $(1, 1), (1, 2), (1, 3)$ give the conditions in this table:

colors	stuffle	shuffle
$(1, 1)$	$A + A^\sigma = 0$	$G(X + Y, X) + G(X + Y, Y) = 0$
$(1, 2)$	$K + V^\sigma = 0$	$A(X + Y, X) - V(X + Y, Y) = 0$
$(1, 3)$	$M - M^\sigma = 0$	$K(X + Y, Y) - K(X + Y, X) = 0$

For the $(1, 2)$ shuffle we have exchanged the two product factors; this merely transposes X, Y and retains exactly the same collection of coefficient rows. Substituting $V = -K^\sigma$ into its shuffle gives $K(X, Y) = -A(Y, Y - X)$. The other two shuffle equations yield the two reflection equations in (2.12).

There are no omitted nonzero color classes. Up to swapping the product factors and conjugation, all unordered pairs are

$$(0, 0), (0, 1), (0, 2), (1, 1), (1, 2), (1, 3), (2, 2).$$

The classes $(0, 0), (0, 2), (2, 2)$ are entirely real and their imaginary rows vanish. The $(0, 1)$ class is exactly Lemma 2.2. This proves necessity and sufficiency. $\square$

2.3 The two independent polynomial sectors

Theorem 2.4 (Canonical description of the full nullspace). *For even w , $\ker A_w = 0$. For odd $w = 2m + 1$, its elements are in one-to-one correspondence with independent choices of degree $n = 2m - 1$ homogeneous polynomials G, A satisfying*

$$G(X, Y) = -G(X, X - Y), \quad A(X, Y) = -A(Y, X). \quad (2.13)$$

All six polynomials are then given by

$$U(X, Y) = -G(Y, X) + cY^n, \quad c = [X^n]G, \quad (2.14)$$

$$M(X, Y) = G(X - Y, X), \quad (2.15)$$

$$K(X, Y) = -A(Y, Y - X), \quad (2.16)$$

$$V(X, Y) = -K(Y, X). \quad (2.17)$$

Each sector has dimension m . In particular, $\dim \ker A_w = 2m = w - 1$ at odd weight.

Proof. The preceding lemmas show that every kernel element is of the form (2.17), with the two conditions (2.13). It remains to impose the symmetry of M and the reflection symmetry of K . Homogeneity and the reflection condition on G give

$$\begin{aligned} M(Y, X) &= G(Y - X, Y) = (-1)^n G(X - Y, -Y) \\ &= -(-1)^n G(X - Y, X) = (-1)^{n+1} M(X, Y). \end{aligned}$$

Likewise, antisymmetry of A gives

$$\begin{aligned} K(X, X - Y) &= -A(X - Y, -Y) \\ &= -(-1)^n A(Y - X, Y) = (-1)^n A(Y, Y - X) \\ &= (-1)^{n+1} K(X, Y). \end{aligned}$$

If n is even, the required equalities $M = M^\sigma$ and $K(X, Y) = K(X, X - Y)$ force $M = K = 0$, hence $G = A = 0$, and then all six polynomials vanish. If n is odd, these two remaining equalities hold automatically. The exhaustive list of classes in Lemma 2.3 proves that every pair in (2.13) is a kernel element, so this is a bijective parameterization.

For the G sector, change variables from X, Y to $X, Z = 2Y - X$. Reflection changes Z to $-Z$, so G is an odd polynomial in Z . At total degree $2m - 1$, a basis is

$$G_j(X, Y) = X^{2j}(2Y - X)^{2m-1-2j}, \quad 0 \leq j < m. \quad (2.18)$$

For the A sector, a basis is

$$A_j(X, Y) = X^{2m-1-j}Y^j - X^jY^{2m-1-j}, \quad 0 \leq j < m. \quad (2.19)$$

Their independence follows respectively from the distinct powers of Z , and from their disjoint exchanged monomial pairs. Thus each sector has dimension m . $\square$

Proof of Theorem 2.1. The full matrix has $6w - 7$ columns. Subtracting the kernel dimensions in Theorem 2.4 gives (2.4). To pass to A_w^g , extend a vector on the remaining columns by zero on the deleted Gaussian columns. It is in $\ker A_w^g$ if and only if the extended vector is in $\ker A_w$ and $G = 0$. At odd weight this leaves exactly the m -dimensional A sector; at even weight it leaves zero. There are $(6w - 7) - (w - 1) = 5w - 6$ remaining columns. Subtracting these kernel dimensions gives (2.5). $\square$

The Hilbert series of these formal nullspaces are especially simple:

$$\sum_{w \geq 2} \dim \ker A_w t^w = \frac{2t^3}{(1-t^2)^2}, \quad \sum_{w \geq 2} \dim \ker A_w^{-g} t^w = \frac{t^3}{(1-t^2)^2}.$$

The second is the previously obtained quotient count; the first records the additional freely liftable Gaussian sector.

2.4 Every Gaussian-only consequence is already a same-point shuffle

Let $\mathcal{R}_w$ be the row space of A_w and let E_g be the coordinate subspace supported only on the Gaussian columns. Projection of $\mathcal{R}_w$ onto the other columns has dimension $\text{rank } A_w^{-g}$. Hence

$$\dim(\mathcal{R}_w \cap E_g) = \text{rank } A_w - \text{rank } A_w^{-g} = \begin{cases} w-1, & w \text{ even,} \\ (w-1)/2, & w \text{ odd.} \end{cases} \quad (2.20)$$

Corollary 2.5 (Completeness for Gaussian-only coefficient rows). *For odd $w = 2m + 1$, every rational linear combination of the defining rows whose non-Gaussian coefficients cancel is a rational linear combination of the m same-point product-shuffle rows obtained from*

$$\text{Li}_p(i) \text{Li}_{w-p}(i), \quad 1 \leq p \leq m.$$

Thus enlarging a search to all the colors and one-divergence rows in A_w yields no additional Gaussian-only coefficient relation in odd weight beyond these elementary shuffles.

Proof. These m rows are present in A_w and are supported only on G . They are independent: in the usual inner-index ordering, the row with $p \leq m$ has largest inner index $q = w - p$, with coefficient one, and no larger inner index occurs. Ordering the selected rows and columns accordingly gives a triangular square block with unit diagonal. Their span is therefore an m -dimensional subspace of $\mathcal{R}_w \cap E_g$, which also has dimension m by (2.20). The spaces are equal. $\square$

This is a statement about the coefficient projection of the equations. When actual depth-one right sides are restored, any equality of row coefficients also gives the corresponding identity of those right sides, because all the underlying analytic rows are valid. The corollary does not forbid Gaussian-only identities from other functional equations or higher-depth elimination.

2.5 A complete membership test and exact certificates

Equations (2.18)–(2.19), followed by (2.17), give $w - 1$ explicit integer vectors $v_1, \dots, v_{w-1}$ in odd weight. For any rational coefficient row t , elementary finite-dimensional duality gives

$$t \in \text{rowspan}_{\mathbb{Q}} A_w \iff t \cdot v_j = 0 \quad \text{for every } j. \quad (2.21)$$

Thus membership in the stated coefficient module has a complete test using only integer polynomial substitution and rational dot products. A nonzero dot product supplies a short exact separating functional. At even weight the row space is the whole coordinate space. None of these tests asserts the absence of a relation outside this matrix.

At weight five, $n = 3$, one may choose

$$G_0 = (2Y - X)^3, \quad G_1 = X^2(2Y - X), \quad A_0 = X^3 - Y^3, \quad A_1 = X^2Y - XY^2.$$

These four parameters give the four-dimensional nullspace of the 92×23 matrix, with no numerical polylogarithm computation. The G -zero subspace has the two A parameters,

explaining the rank pair (19, 17). At weight seven the corresponding pair is (29, 26), and at weight thirty-one it is (149, 134).

The supplied standard-library program `full_gaussian_rank.py` constructs the raw rows from (2.3); its separate polynomial code constructs (2.17). It verifies that every displayed basis vector annihilates every raw row. Independent rational elimination checks both ranks in weights 2 through 11. Basis identities are also replayed in weights 12 through 17 and in weights 23, 31, 47, 63. The JSON receipt includes the entire weight-five matrix, coordinate order, row labels, and kernel basis. These finite checks validate the implementation; the preceding proof establishes all weights.

2.6 Integration notes and further questions

The manuscript's Conjecture `signed:conj:rank` can be replaced by Theorem 2.1. The even-weight formulas and the canonical kernel description are additional results. The word “equivalently” preceding the manuscript's row-space-intersection formulation should be changed to “consequently,” unless one of the absolute rank formulas is taken as an additional premise: a specified difference of two ranks does not by itself determine both ranks. Our proof establishes the absolute ranks and their difference.

Three directions now have precise formulations. First, replace the implicit numerical rank routine by the explicit polynomial test (2.21) in future relation searches. Second, add specified distribution or octahedral rows and compute their action on these two small polynomial sectors, rather than repeating elimination in all $6w - 7$ coordinates. Third, classify which higher-depth relations couple the two sectors. Such a classification would address a larger relation module; passing from any formal module to numerical period independence remains a separate problem.

3 A universal splitting at every cyclotomic level

The boundary lift used above does not depend on the number four. It separates the same-point coordinates from the remaining color problem at every cyclotomic level.

Let $N \geq 3$, $\omega = e^{2\pi i/N}$, and define the formal imaginary coordinates for $D_{a,w-a}(\omega^r, \omega^s)$ exactly as before, now with colors modulo N . Impose conjugation, discard its fixed coordinates, and remove one divergent coordinate $X_1^{0,r}$ for each conjugate pair of nonreal roots. Let $A_{N,w}$ be the matrix of finite single-product shuffle and stuffle rows, with only their difference retained when exactly one factor diverges. Put

$$R_N = \{1, \dots, \lfloor (N-1)/2 \rfloor\}, \quad k_N = |R_N|.$$

The matrix has

$$d_{N,w} = \frac{N^2 - \gcd(N, 2)^2}{2}(w-1) - k_N$$

columns: conjugation has $\gcd(N, 2)^2$ fixed color pairs, and the last term counts the removed divergent coordinates. Let $A_{N,w}^{-g}$ delete all $k_N(w-1)$ same-point columns $X_a^{r,0}$ with $r \in R_N$. These are formal matrices with the same restricted meaning as A_w ; distribution and higher-depth rows have not been added.

For a homogeneous degree- n polynomial, where $n = w - 2$, define

$$\mathcal{H}_n^- = \{G \in \mathbb{Q}[X, Y]_n : G(X, Y) = -G(X, X - Y)\}.$$

Theorem 3.1 (Universal same-point splitting). *For odd $w = 2m + 1$, there is a canonical split exact sequence*

$$0 \longrightarrow \ker A_{N,w}^{-g} \longrightarrow \ker A_{N,w} \longrightarrow \bigoplus_{r \in R_N} \mathcal{H}_{2m-1}^- \longrightarrow 0.$$

The last map records the same-point polynomials G_r . In particular,

$$\dim \ker A_{N,w} - \dim \ker A_{N,w}^{-g} = k_N m. \quad (3.1)$$

At even w , every vector in $\ker A_{N,w}$ has all same-point coordinates zero, so the two kernels are naturally identical. Consequently

$$\text{rank } A_{N,w} - \text{rank } A_{N,w}^{-g} = \begin{cases} k_N(w-1), & w \text{ even}, \\ k_N(w-1)/2, & w \text{ odd}. \end{cases} \quad (3.2)$$

Proof. Fix $r \in R_N$ and abbreviate the polynomials for colors $(r, 0), (0, r), (r, -r)$ by G_r, U_r, M_r . The product-color classes $(r, r), (0, r), (r, -r)$ imply exactly the same necessary conditions as in the Gaussian proof:

$$\begin{aligned} G_r(X, Y) &= -G_r(X, X - Y), \\ U_r(X, Y) &= -G_r(Y, X) + c_r Y^n, \quad c_r = [X^n]G_r, \\ M_r(X, Y) &= G_r(X - Y, X), \quad M_r = M_r^\sigma. \end{aligned}$$

The boundary equation uses the single regularized difference, so its derivation is unchanged. Homogeneity gives $M_r^\sigma = (-1)^{n+1}M_r$. Thus even n forces $G_r = 0$. For odd n , the symmetry of M_r is automatic.

For odd n , define a section by assigning the displayed values to these three color pairs for each r , their negatives to conjugate pairs, and zero to every other color pair. This assignment satisfies all rows. Here is a support check that prevents an implicit assumption about the other colors. A shuffle can meet a nonzero G_r coordinate only when its two product colors are (r, r) , and then it is exactly the antireflection equation. It can meet U_r or M_r only for product colors $(0, r)$ or $(r, 0)$, and then it is exactly the boundary equation. The stuffle rows meeting these coordinates are the boundary class and $(r, -r)$, which is the symmetry of M_r . Transposition and conjugation cover the reversed and conjugate classes. Thus no unexamined row couples the proposed section to other colors. The supports for distinct $r \in R_N$ are disjoint.

Subtracting this section from any kernel vector leaves all $G_r = 0$; the remainder is naturally a vector in $\ker A_{N,w}^{-g}$. This proves the splitting. Since $\dim \mathcal{H}_{2m-1}^- = m$, it gives (3.1). Subtracting kernel dimensions from column counts proves (3.2). $\square$

At odd weight, an immediate consequence is that every coefficient relation supported only on these same-point columns is spanned by the independent same-point shuffle rows, separately for each root pair. Those rows already supply $k_N(w-1)/2$ dimensions, equal to (3.2). Thus the Gaussian-only completeness result above is one instance of a level-independent fact about this particular single-product coefficient system.

3.1 The complete level-three ranks

The earlier incoming report [2] already identifies the residual level-three space after $G = 0$. In that case the boundary formulas force $U = M = 0$, so its regularized presentation and the present removed-coordinate presentation agree. Its remaining polynomial satisfies

$$A(Y, X) = -A(X, Y), \quad A(X, X - Y) = A(X, Y),$$

and is parameterized by

$$A = \Delta \mathbb{Q}[Q, \Delta^2], \quad \Delta = XY(X - Y), \quad Q = X^2 - XY + Y^2.$$

At odd weight its homogeneous degree- $w - 2$ component has dimension $\lfloor (w + 1)/6 \rfloor$, and at even weight it is zero. For completeness, antisymmetry and reflection force the three linear factors of Δ ; after their removal the quotient is invariant under the group generated by the two transformations. That group acts as permutations and global negation on $(X, -Y, Y - X)$, whose sum is zero, giving the invariant ring $\mathbb{Q}[Q, \Delta^2]$. Counting monomials of degrees $3 + 2a + 6b = w - 2$ gives the stated dimension. This residual description is credited to the preceding report; the new assertion is its extension to the free same-point family under the manuscript's boundary convention.

Corollary 3.2 (Level-three extension). *The analogous full level-three matrix has $4w - 5$ columns and satisfies*

$$\text{rank } A_{3,w} = \begin{cases} 4w - 5, & w \text{ even}, \\ (7w - 9)/2 - \lfloor (w + 1)/6 \rfloor, & w \text{ odd}, \end{cases}$$

while the matrix with all same-point columns deleted satisfies

$$\text{rank } A_{3,w}^{-g} = \begin{cases} 3w - 4, & w \text{ even}, \\ 3w - 4 - \lfloor (w + 1)/6 \rfloor, & w \text{ odd}. \end{cases}$$

Proof. At odd weight the full kernel has dimension $(w - 1)/2 + \lfloor (w + 1)/6 \rfloor$ by Theorem 3.1; its restricted kernel has only the second summand. Both vanish at even weight. Subtraction from the respective column counts proves the formulas. $\square$

The supplementary program `universal_gaussian_lift.py` verifies the lift directly against every raw row at selected weights and levels 3, 4, 5, 6, 8, 10, 12. Independent exact ranks in weights 2, 3, 4 at levels 3 through 6 check the rank differences. This establishes neither a formula for the remaining kernel at arbitrary level nor numerical period independence; those remain separate questions.

4 A complete formal classification of rational companion values

Put

$$J(x) = \int_0^1 \frac{\log(1 + t^x)}{1 + t} dt, \quad F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}.$$

Radchenko–Zagier's identity [7] is

$$J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}. \quad (4.1)$$

All identities in this section at positive real arguments use real logarithms. Our negative statements concern the rationalized five-term calculus, not numerical independence of periods.

Write $A(K) = K^\times \otimes_{\mathbb{Z}} \mathbb{Q}$ additively, using $\langle u \rangle$ for the class of u . Roots of unity disappear. Let $\mathcal{P}(K)$ be the rationalized pre-Bloch group, with $\partial[z] = \langle z \rangle \wedge \langle 1 - z \rangle$ and $[0] = [1] = 0$. For coprime positive p, q , let $\xi_{p/q}$ denote the Galois-invariant element supplied by the finite cyclotomic formula for $F(p/q) - F(1)$. Its normalization is

$$\partial \xi_{p/q} = \beta_q(p) - \beta_p(q) - \langle p \rangle \wedge \langle q \rangle, \quad \beta_m(a) = \sum_{k=1}^{m-1} \langle 1 - \zeta_m^{ak} \rangle \wedge \langle 1 - \zeta_m^k \rangle. \quad (4.2)$$

Set $\beta_1 = \beta_2 = 0$. Fractions in subscripts are always reduced before using (4.2). Define

$$\eta_{p/q} = \xi_{2p/q} - 2\xi_{p/q} + \xi_{p/(2q)} \quad \text{in } \mathcal{P}(\mathbb{Q}(\zeta_{2pq})). \quad (4.3)$$

A formal logarithmic reduction of $J(p/q)$ means $\eta_{p/q} = 0$; a formal rational-dilogarithm reduction means $\eta_{p/q} \in \text{im } \mathcal{P}(\mathbb{Q})$. Products of algebraic logarithms and rational multiples of π^2 are the terms suppressed by this terminology.

We use the established rationalized facts

$$\ker(\partial|_{\mathcal{P}(K)})^{\text{Gal}(K/\mathbb{Q})} = 0, \quad \partial\mathcal{P}(\mathbb{Q}) = \Lambda^2 A(\mathbb{Q})$$

for Galois number fields K [1, `hstruct:eq:bloch-facts`]. These are the Bloch-group descent and $K_2(\mathbb{Q})$ inputs already used in the manuscript. Consequently, for an invariant η , the two kinds of reduction are equivalent, respectively, to $\partial\eta = 0$ and $\partial\eta \in \Lambda^2 A(\mathbb{Q})$.

4.1 The existing Gram theorem and a new subfield consequence

For $M \geq 3$, set $G_M = (\mathbb{Z}/M\mathbb{Z})^\times / \{\pm 1\}$. The all-conductor Gram theorem of the manuscript [1, `hstruct:thm:kernel`, `hstruct:lem:positive-Gram`] states that

$$\sum_{a \in G_M} c_a \beta_M(a) = 0 \iff c_a = c_{a^{-1}} \quad (a \in G_M). \quad (4.4)$$

In particular $\beta_M(a) = 0$ precisely when $a^2 \equiv \pm 1 \pmod{M}$. Its proof provides more information than this zero test. Put

$$v_k = (\log |1 - \zeta_M^{bk}|)_{b \in G_M}, \quad A_M = \sum_{k=1}^{M-1} v_k v_k^\top, \quad (P_a v)_b = v_{ba}.$$

Then A_M is positive definite, commutes with every P_a , and the logarithmic exterior image of $\beta_M(a)$ is

$$A_M(P_a - P_{a^{-1}}). \quad (4.5)$$

The positivity uses all roots, including roots of lower order: for a nontrivial even character of primitive conductor $f \mid M$, the $k = M/f$ vector has a nonzero Fourier coefficient proportional to $\tau(\chi)L(1, \bar{\chi})$. Prime-order roots supply the trivial character. This is an exact nonvanishing argument, not a numerical determinant test. We never assume that the logarithmic exterior map is injective on the whole exterior square.

Theorem 4.1 (No proper real subfield for a nonzero individual symbol). *Let $L = \mathbb{Q}(\zeta_M)^+$ and let E be a proper subfield of L . If the element $\beta_M(a)$ belongs to the image of $\Lambda^2 A(E)$ in $\Lambda^2 A(L)$, then $\beta_M(a) = 0$.*

Proof. The symbols do belong to the real field, because

$$\langle 1 - \zeta_M^k \rangle = \tfrac{1}{2} \langle (1 - \zeta_M^k)(1 - \zeta_M^{-k}) \rangle.$$

Let $H = \text{Gal}(L/E)$, a nontrivial subgroup of G_M . Logarithmic vectors of elements of E lie in $W = (\mathbb{R}^{G_M})^H$. A skew matrix arising from $\Lambda^2 W$ annihilates $W^\perp$. On the complex character line of a character χ nontrivial on H , formula (4.5) therefore gives

$$\lambda_\chi(\chi(a) - \chi(a)^{-1}) = 0, \quad \lambda_\chi > 0.$$

Thus $\chi(a)^2 = 1$ for every such character. Let X_0 be the proper subgroup of characters trivial on H , and choose $\chi_0 \notin X_0$. For every $\psi \in X_0$, both χ_0 and $\chi_0\psi$ are outside X_0 . Dividing their squared values gives $\psi(a)^2 = 1$. Thus all characters of G_M are trivial at a^2 . Characters separate points, so $a^2 = 1$ in G_M , and (4.4) gives $\beta_M(a) = 0$. $\square$

This statement concerns a single $\beta_M(a)$. It does not assert that every nonzero linear combination of symbols has this property.

4.2 Two exact dyadic identities

Lemma 4.2 (Odd conductor doubling). *If m is odd and a is odd with $(a, m) = 1$, then, inside $\mathbb{Q}(\zeta_{2m}) = \mathbb{Q}(\zeta_m)$,*

$$\beta_{2m}(a) = 3\beta_m(a) - \beta_m(2a) - \beta_m(a/2), \quad (4.6)$$

where division by 2 means multiplication by its inverse modulo m .

Proof. The even-indexed roots contribute $\beta_m(a)$. Write the odd-indexed roots as $-\zeta_m^j$, $0 \leq j < m$. The $j = 0$ term is $\langle 2 \rangle \wedge \langle 2 \rangle = 0$. For $j \neq 0$, put $u_j = \langle 1 - \zeta_m^j \rangle$. Factorization gives $\langle 1 + \zeta_m^j \rangle = u_{2j} - u_j$. The remaining contribution is

$$\sum_{j=1}^{m-1} (u_{2aj} - u_{aj}) \wedge (u_{2j} - u_j) = 2\beta_m(a) - \beta_m(2a) - \beta_m(a/2).$$

Adding the even-root contribution proves the identity. $\square$

Lemma 4.3 (Even conductor second difference). *Let $e \geq 2$ be even and $(a, 2e) = 1$. Put*

$$\Delta_e(a) = \beta_{2e}(a) - 2\beta_e(a) + \beta_{e/2}(a).$$

Then

$$\Delta_e(a) = 0 \iff a^2 \equiv 1 \pmod{2e}. \quad (4.7)$$

Proof. For $e = 2$ all three symbols vanish and every odd square is 1 modulo 4. Suppose $e \geq 4$. Let N be the normalized field norm from $\mathbb{Q}(\zeta_{2e})$ to $\mathbb{Q}(\zeta_e)$, acting on A by division by the extension degree 2. Even-indexed roots are fixed. If k is odd, then

$$N\langle 1 - \zeta_{2e}^k \rangle = \frac{1}{2}\langle 1 - \zeta_e^k \rangle,$$

because the nontrivial automorphism sends ζ_{2e} to $-\zeta_{2e}$. Every odd residue modulo e has two odd lifts modulo $2e$, hence

$$(\Lambda^2 N)\beta_{2e}(a) = \beta_e(a) + \frac{1}{2}(\beta_e(a) - \beta_{e/2}(a)). \quad (4.8)$$

It follows that

$$(\Lambda^2 N)\Delta_e(a) = -\frac{1}{2}(\beta_e(a) - \beta_{e/2}(a)).$$

If $\Delta_e(a) = 0$, then $\beta_e(a) = \beta_{e/2}(a)$, and its definition gives $\beta_{2e}(a) = \beta_e(a)$. The right side belongs to $\Lambda^2 A(\mathbb{Q}(\zeta_e)^+)$. This real field has index 2 in $\mathbb{Q}(\zeta_{2e})^+$, so Theorem 4.1 implies $\beta_{2e}(a) = 0$. Thus $a^2 \equiv \pm 1 \pmod{2e}$. Since $4 \mid 2e$ and a is odd, the negative sign is impossible. Conversely, $a^2 \equiv 1 \pmod{2e}$ makes each of the three symbols zero. $\square$

4.3 The two odd-conductor coefficient tests

For odd m and $(a, m) = 1$, define

$$D_m(a) = \beta_m(a) - \beta_m(a/2), \quad T_m(a) = \beta_m(2a) - 2\beta_m(a) + \beta_m(a/2).$$

The convention at $m = 1$ is zero.

Lemma 4.4. *For every odd $m \geq 1$ and every unit a modulo m ,*

$$D_m(a) = 0 \iff m \in \{1, 3, 5\}, \quad (4.9)$$

$$T_m(a) = 0 \iff a^2 \equiv \pm 1 \pmod{m}. \quad (4.10)$$

Proof. By (4.4), equality of two symbols $\beta_m(a) = \beta_m(b)$ occurs only when $a = b$ in G_m or when both classes are self-inverse. Take $b = a/2$. The first alternative gives $2 = 1$ in G_m , hence $m = 1$ or 3 . In the second, both a^2 and $(a/2)^2$ are signs, so 4 is a sign modulo m ; thus $m = 1, 3, 5$. Conversely, the symbol is identically zero at these three odd conductors.

For the second assertion, use the group algebra $\mathbb{R}[G_m]$. Let g be the class of 2 and let $V_a = [a] - [a^{-1}]$. The image of $T_m(a)$ under the exact coefficient model is

$$(P_g + P_{g^{-1}} - 2I)V_a.$$

On each orbit of multiplication by g , the kernel of this cyclic Laplacian consists of constants. If $V_a \neq 0$, its support consists of two points with opposite coefficients. It cannot be constant on nontrivial g -orbits. If $g = 1$, then $m = 1$ or 3 and $V_a = 0$ anyway. Thus $T_m(a) = 0$ forces $V_a = 0$, which is precisely the square congruence. The converse follows from the same displayed formula. $\square$

4.4 Separation and classification

If r, s are coprime, normalized norms on the individual exterior factors separate symbols from $\mathbb{Q}(\zeta_r)$ and $\mathbb{Q}(\zeta_s)$. In detail, every summand of a β -symbol has conjugate factors. Its normalized norm to $\mathbb{Q}$ has equal factors and therefore zero wedge. A normalized norm from the compositum to one of the two fields fixes symbols in that field and kills symbols from the other, by linear disjointness. Consequently, for combinations U_r, U_s of β -symbols at divisors of r, s ,

$$U_r - U_s \in \Lambda^2 A(\mathbb{Q}) \iff U_r = U_s = 0. \quad (4.11)$$

Indeed, taking either field norm and then the norm to $\mathbb{Q}$ first shows that the rational wedge must be zero, and then that both components vanish. This uses norms on factors; averaging a Galois-invariant exterior symbol itself would not give separation.

Theorem 4.5 (All rational values of the companion). *Let p, q be coprime positive integers.*

1. *If p and q are odd, $J(p/q)$ has a formal rational-dilogarithm reduction exactly when $p, q \in \{1, 3, 5\}$. It has a formal logarithmic reduction exactly when $p = q = 1$.*
2. *Otherwise, let e be the even member of $\{p, q\}$ and o the odd member. Formal rational-dilogarithm and formal logarithmic reductions are equivalent, and occur exactly when*

$$e^2 \equiv \pm 1 \pmod{o}, \quad o^2 \equiv 1 \pmod{2e}. \quad (4.12)$$

Congruences modulo 1 are vacuous.

Proof. If p, q are odd, equations (4.2) and (4.6) give the exact boundary

$$\partial\eta_{p/q} = D_q(p) - D_p(q) + \langle 2 \rangle \wedge \langle p/q \rangle. \quad (4.13)$$

The field separation (4.11) and (4.9) show that this boundary is rational precisely when $p, q \in \{1, 3, 5\}$. The normalized norm to $\mathbb{Q}$ sends the boundary to $\langle 2 \rangle \wedge \langle p/q \rangle$. Since p/q has odd numerator and denominator, its prime-exponent vector is independent of that of 2 unless $p = q = 1$. Thus the boundary is zero only in that case.

Now suppose $p = e$ is even and $q = o$ is odd. The argument $x/2$ reduces to $(e/2)/o$, and the rational wedge terms cancel exactly. Hence

$$\partial\eta_{e/o} = T_o(e) - \Delta_e(o). \quad (4.14)$$

The fields with conductors o and $2e$ are linearly disjoint. By (4.11), a rational boundary is possible exactly when both blocks vanish; it is then zero. Lemmas 4.3 and 4.4 give precisely (4.12).

Finally, $\xi_x + \xi_{1/x} = 0$ in the invariant pre-Bloch group: their boundaries cancel, and invariant Bloch descent applies. Pairing the three terms in (4.3) gives $\eta_x + \eta_{1/x} = 0$. This handles odd numerator and even denominator. In each positive case, invariant Bloch descent also proves the asserted converse from the boundary test to the actual formal reduction. $\square$

Remark 4.6 (Why the doubled modulus cannot be dropped). The pair $(e, o) = (8, 3)$ satisfies $e^2 \equiv 1 \pmod{o}$ and $o^2 \equiv 1 \pmod{e}$, but $o^2 = 9 \not\equiv 1 \pmod{16}$. Thus $F(3/8) - F(1)$ has the manuscript's rational-dilogarithm core, while $J(3/8)$ fails the formal rational-dilogarithm test. Similarly $(10, 3)$ passes the two signed congruences for F but fails the condition for J . These are exact algebraic obstructions, not failed integer-relation searches.

Corollary 4.7 (Two infinite families). *For every integer $n \geq 1$, $J(n/(n+1))$ has a formal logarithmic reduction. For every even integer $e \geq 2$, both $J(e/(e^2-1))$ and $J(e/(e^2+1))$ do as well.*

Proof. For consecutive integers, write the odd one as $o = e \pm 1$. Then $e^2 \equiv 1 \pmod{o}$ and $o^2 - 1 = e(e \pm 2)$ is divisible by $2e$. For $o = e^2 \pm 1$, the first condition is immediate and $o^2 - 1 = e^2(e^2 \pm 2)$ is divisible by $2e$ because e is even. $\square$

5 The exact rank of the denominator obstruction

The single-value criterion also yields a complete finite-dimensional calculation for the denominator part of the companion. Let q be odd, $G = G_q$, and $H = \langle 2 \rangle \subseteq G$. For a finite abelian group U , write

$$d(U) = \frac{|U| - |U[2]|}{2}.$$

The integer $d(U)$ is the dimension of the subspace of $\mathbb{Q}[U]$ that changes sign under inversion. For the boundary case $q = 1$, use $G_1 = H_1 = \{1\}$ and $\rho(1) = 0$; the symbol families are zero, so the assertions below are vacuous there.

Theorem 5.1 (Rank and all coefficient relations). *The two subspaces*

$$\text{span}_{\mathbb{Q}}\{D_q(a) : a \in G\}, \quad \text{span}_{\mathbb{Q}}\{T_q(a) : a \in G\}$$

are equal. Their common dimension is

$$\boxed{\rho(q) = d(G) - d(G/H) = \frac{|G| - |G[2]| - |G/H| + |(G/H)[2]|}{2}}. \quad (5.1)$$

For rational coefficients c_a , the exact relation test for the second-difference family is

$$\sum_{a \in G} c_a T_q(a) = 0 \iff c_a - c_{a^{-1}} \text{ is constant on each coset of } H. \quad (5.2)$$

For the first-difference family the exact relation test is

$$\sum_{a \in G} c_a D_q(a) = 0 \iff c_a - c_{2a} = c_{a^{-1}} - c_{2a^{-1}} \quad (a \in G). \quad (5.3)$$

Proof. The existing all-conductor kernel theorem identifies the symbol span with the inversion-odd subspace V^- of $\mathbb{Q}[G]$, by $\beta_q(a) \mapsto [a] - [a^{-1}]$. Work first over $\mathbb{R}$, with its usual inner product, and write $P_g[a] = [ga]$, where g is the class of 2. The cyclic Laplacian $L_g = P_g + P_{g^{-1}} - 2I$ is self-adjoint, preserves V^- , and has kernel the functions constant on H -cosets. Thus the span of the $T_q(a)$ is the orthogonal complement in V^- of $(V^-)^H$. The latter has dimension $d(G/H)$, proving (5.1) for T and the coefficient test (5.2).

An inversion-odd vector w is orthogonal to every $D_q(a) = ([a] - [a^{-1}]) - ([a/g] - [ga^{-1}])$ precisely when

$$2(w_a - w_{a/g}) = 0 \quad (a \in G).$$

This is exactly the condition that w be constant on H -cosets. Hence the two spans have the same orthogonal complement and are equal. All matrices have rational entries, so these dimensions and equalities descend to $\mathbb{Q}$. Finally, the coefficient of $[a]$ in the inversion-odd image of a D -combination is $c_a - c_{ga} - c_{a^{-1}} + c_{ga^{-1}}$, proving (5.3). $\square$

For example, at $q = 31$, G is cyclic of order 15 and the class of 2 has order 5. Therefore

$$d(G) = 7, \quad d(G/H) = d(C_3) = 1, \quad \rho(31) = 6.$$

The loss of one dimension is an exact distribution relation, not a numerically small singular value. The relations include the evident telescoping identities

$$\sum_{h \in H} D_q(ah) = 0, \quad \sum_{h \in H} T_q(ah) = 0,$$

and the theorem describes all additional consequences of inversion.

Corollary 5.2 (A certified lower bound for formal companion values). *For each odd q , there exist $\rho(q)$ positive even numerators e_j , coprime to q , such that the formal elements $\eta_{e_j/q}$ are linearly independent modulo rational dilogarithms.*

Proof. Choose residues giving a basis among $T_q(a)$, and choose even positive representatives e_j of those residues. In a common cyclotomic field, apply the exterior square of the normalized norm to $\mathbb{Q}(\zeta_q)$. It kills every numerator-conductor block and keeps the corresponding $T_q(e_j)$. If a combination of the $\eta_{e_j/q}$ belonged to the rational pre-Bloch group, this projection would be a rational wedge. The normalized norm to $\mathbb{Q}$ kills the symbol combination and fixes that wedge, so the wedge is zero. The chosen T symbols are independent, proving the assertion. $\square$

6 Explicit logarithmic evaluations and a rational dilogarithm value

The formal classification is complemented by full analytic evaluations. These proofs use the positive real domain throughout, so their logarithmic correction terms and multiples of π^2 are fixed exactly.

Define, as in Radchenko–Zagier [7],

$$\mathcal{F}(x) = F(x) - F(1) + \frac{\pi^2}{12}(x - 2 + x^{-1}) - \frac{1}{4} \log^2 x, \quad \mathcal{J}(x) = J(x) - \frac{1}{2} \log^2 2 + \frac{\pi^2}{24}(x - x^{-1}).$$

Let $L(z) = \text{Li}_2(z) + \frac{1}{2} \log z \log(1 - z)$ for $0 < z < 1$, with the continuous endpoint values. The

following are established identities from Radchenko–Zagier, equations (15) and (22a) [7]:

$$\mathcal{F}(x) - \mathcal{F}(x+1) - \mathcal{F}\left(\frac{x}{x+1}\right) = L(1/2) - L\left(\frac{x}{x+1}\right), \quad (6.1)$$

$$\mathcal{F}(x) + \mathcal{F}(1/x) = 0, \quad (6.2)$$

$$\mathcal{F}(2x) + \mathcal{F}(x/2) + \mathcal{F}\left(\frac{x+1}{2}\right) + \mathcal{F}\left(\frac{2x}{x+1}\right) - 3\mathcal{F}(x) = L\left(\frac{x}{x+1}\right) - L(1/2) + \frac{1}{2} \log 2 \log x. \quad (6.3)$$

We use them as established analytic inputs, not as new results. The Rogers five-term relation on $(0, 1)^2$ is

$$L(u) + L(v) = L(uv) + L\left(\frac{u(1-v)}{1-uv}\right) + L\left(\frac{v(1-u)}{1-uv}\right). \quad (6.4)$$

Unlike its continuation to arbitrary real arguments, this displayed identity holds as an equality in $\mathbb{R}$, with no unspecified multiple of π^2 .

The next lemma is a useful reformulation of known functional equations: it also follows from Choie–Kumar’s Theorem 3.2(1), equation (3.3), after reciprocity and the three-term equation for $\mathcal{F}$ [8]. We give a direct positive-real proof to fix every correction term. We do not claim it as a new general functional equation. The explicit rational specializations below are proposed additions to the manuscript; no exhaustive claim of historical priority is made for those evaluations.

Lemma 6.1 (A finite difference at adjacent positive arguments). *For every real $x > 0$,*

$$\begin{aligned} \mathcal{J}\left(\frac{x}{x+1}\right) &= \mathcal{F}(x) - \mathcal{F}(x+1) - \mathcal{F}(2x) + \mathcal{F}(2x+2) - L(1/2) \\ &\quad + \frac{1}{2} \log 2 \log \left(\frac{x}{x+1}\right). \end{aligned} \quad (6.5)$$

Proof. Let $r = x/(x+1)$. By (6.3) and $\mathcal{J}(r) = \mathcal{F}(2r) - 2\mathcal{F}(r) + \mathcal{F}(r/2)$,

$$\mathcal{J}(r) = \mathcal{F}(r) - \mathcal{F}\left(\frac{2x+1}{2x+2}\right) - \mathcal{F}\left(\frac{2x}{2x+1}\right) + L\left(\frac{x}{2x+1}\right) - L(1/2) + \frac{1}{2} \log 2 \log r.$$

Apply (6.1) at $2x$ and $2x+1$. The two middle $\mathcal{F}$ terms telescope to

$$-\mathcal{F}(2x) + \mathcal{F}(2x+2) + 2L(1/2) - L\left(\frac{2x}{2x+1}\right) - L\left(\frac{2x+1}{2x+2}\right).$$

In (6.4), take $u = 2x/(2x+1)$ and $v = (2x+1)/(2x+2)$. The three arguments on the right are r , $x/(2x+1)$, and $1/2$. Thus all Rogers terms in the preceding expression reduce to $-L(r)$. The result is

$$\mathcal{J}(r) = \mathcal{F}(r) - L(r) - \mathcal{F}(2x) + \mathcal{F}(2x+2) + \frac{1}{2} \log 2 \log r.$$

One final application of (6.1) at x gives (6.5). □

Theorem 6.2 (The complete consecutive rational family). *Put*

$$S(n) = \sum_{k=1}^{n-1} \log^2 \left(2 \sin \frac{\pi k}{n} \right), \quad S(1) = 0.$$

For every integer $n \geq 1$,

$$\boxed{J\left(\frac{n}{n+1}\right) = \frac{1}{2} \log^2 2 - \frac{n^2 - n - 1}{24n(n+1)} \pi^2 + \frac{1}{2} (S(n) - S(n+1) - S(2n) + S(2n+2)).} \quad (6.6)$$

Every logarithm has a positive algebraic argument.

Proof. The established integer value of F [7, eq. (23)] gives

$$\mathcal{F}(n) = \frac{n-1}{24}\pi^2 + \frac{1}{4}\log^2 n + \frac{1}{2}S(n). \quad (6.7)$$

Insert this into (6.5), with $x = n$. The π^2 terms from the four $\mathcal{F}$ values sum to $\pi^2/24$; subtracting $L(1/2) = \pi^2/12$ leaves $-\pi^2/24$. The ordinary log-square terms sum to $\frac{1}{2}\log 2 \log((n+1)/n)$ and cancel the last term of (6.5). Therefore

$$\mathcal{J}\left(\frac{n}{n+1}\right) = -\frac{\pi^2}{24} + \frac{1}{2}(S(n) - S(n+1) - S(2n) + S(2n+2)).$$

Returning from $\mathcal{J}$ to J proves the displayed formula. $\square$

Corollary 6.3 (Two small explicit values). *With $\phi = (1 + \sqrt{5})/2$ and $\varepsilon = 1 + \sqrt{2}$,*

$$J(3/4) = -\frac{5\pi^2}{288} + \frac{1}{8}\log^2 2 + \frac{1}{2}\log^2 \varepsilon, \quad (6.8)$$

$$J(4/5) = -\frac{11\pi^2}{480} + \frac{7}{8}\log^2 2 + 2\log^2 \phi - \frac{1}{2}\log^2 \varepsilon. \quad (6.9)$$

Proof. Direct pairing of the sine factors gives

$$\begin{aligned} S(3) &= \frac{1}{2}\log^2 3, & S(4) &= \frac{3}{2}\log^2 2, \\ S(5) &= \frac{1}{4}\log^2 5 + \log^2 \phi, & S(6) &= \log^2 2 + \frac{1}{2}\log^2 3, \\ S(8) &= \frac{7}{4}\log^2 2 + \log^2 \varepsilon, & S(10) &= \log^2 2 + \frac{1}{4}\log^2 5 + 5\log^2 \phi. \end{aligned}$$

For example, the logarithms of $2\sin(\pi/8)$ and $2\sin(3\pi/8)$ have sum $\frac{1}{2}\log 2$ and difference $\log \varepsilon$. Their squares therefore give the stated $S(8)$ formula. Substitute $n = 3, 4$ in (6.6). $\square$

Theorem 6.4 (A rational-dilogarithm value at $3/5$). *The following analytic identity holds:*

$$\boxed{J(3/5) = \text{Li}_2(1/3) - \frac{1}{2}\text{Li}_2(1/5) - \frac{7\pi^2}{180} + \frac{1}{2}\log^2 2 + \frac{1}{2}\log^2 3 - \frac{1}{4}\log^2 5 + \log^2 \phi.} \quad (6.10)$$

Its rational-dilogarithm core is exactly $[1/3] - \frac{1}{2}[1/5]$ in the formal calculus.

Proof. Take $x = 3/2$ in (6.5) and use (6.2):

$$\mathcal{J}(3/5) = -\mathcal{F}(2/3) + \mathcal{F}(2/5) - \mathcal{F}(3) + \mathcal{F}(5) - L(1/2) + \frac{1}{2}\log 2 \log(3/5).$$

Equation (6.1) at $x = 2$ says $\mathcal{F}(2/3) = \mathcal{F}(2) - \mathcal{F}(3) - L(1/2) + L(2/3)$. Consequently,

$$\mathcal{J}(3/5) = \mathcal{F}(2/5) + \mathcal{F}(5) - \mathcal{F}(2) - L(2/3) + \frac{1}{2}\log 2 \log(3/5). \quad (6.11)$$

Use the established Radchenko–Zagier evaluation [7, eq. (24)]

$$F(2/5) - F(1) = \frac{1}{2}\text{Li}_2(4/5) - \frac{\pi^2}{5} + \log^2\left(2\sin\frac{\pi}{5}\right) + \log^2\left(2\sin\frac{2\pi}{5}\right) + \log 2 \log(2/5),$$

together with (6.7) at 2, 5. The two log-sine squares in this formula have sum $\frac{1}{8}\log^2 5 + \frac{1}{2}\log^2 \phi$. Simplifying (6.11) and applying Euler's reflection identity to $\text{Li}_2(4/5)$ and $\text{Li}_2(2/3)$ gives

$$\mathcal{J}(3/5) = \text{Li}_2(1/3) - \frac{1}{2}\text{Li}_2(1/5) - \frac{\pi^2}{12} + \frac{1}{2}\log^2 3 - \frac{1}{4}\log^2 5 + \log^2 \phi.$$

Since $3/5 - 5/3 = -16/15$, the conversion from $\mathcal{J}$ to J adds $\frac{1}{2} \log^2 2 + 2\pi^2/45$, proving (6.10). All Rogers and Euler reflection arguments lie in $(0, 1)$, so there is no branch ambiguity.

For an independent boundary check,

$$\partial \left([1/3] - \frac{1}{2}[1/5] \right) = \langle 2 \rangle \wedge \langle 3/5 \rangle,$$

which agrees with (4.13) because the symbols at conductors 3 and 5 vanish. $\square$

Proposition 6.5 (Reciprocity and the remaining odd rational cores). *For every $x > 0$,*

$$J(x) + J(1/x) = \log^2 2. \quad (6.12)$$

In particular, the established $J(1/3)$ and $J(1/5)$ formulas yield

$$\begin{aligned} J(3) &= -\frac{\pi^2}{9} + \frac{1}{2} \log^2 2 + \frac{1}{2} \log^2 3 + \text{Li}_2(1/3), \\ J(5) &= -\frac{7\pi^2}{60} + \frac{1}{2} \log^2 2 + \frac{1}{4} \log^2 5 + \log^2 \phi + \frac{1}{2} \text{Li}_2(1/5). \end{aligned}$$

The value $J(5/3)$ is $\log^2 2 - J(3/5)$. These account for all nontrivial odd/odd cases in Theorem 4.5.

Proof. Integration by parts gives

$$J(x) = \log^2 2 - x \int_0^1 \frac{t^{x-1} \log(1+t)}{1+t^x} dt.$$

Substitute $u = t^x$ in the remaining integral to obtain $J(1/x)$. The boundary at zero vanishes and the boundary at one is $\log^2 2$. This reciprocity is elementary and already known; see, for example, Choie–Kumar [8, Theorem 3.1]. It is included to make the analytic consequences complete. The displayed values follow by substitution in the existing $1/3$ and $1/5$ formulas. $\square$

7 Background: the derivative family and its finite zero bound

We recall the exact framework from the zero-geometry chapter of the pinned manuscript [1]. The following statements are established inputs to this continuation; their short proofs are included to make the new endpoint results readable independently of the repository.

The generalized Stieltjes constants use the convention

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n, \quad a > 0.$$

For $0 \leq \rho < 1$, define

$$\ell_n(\rho, a) = \sum_{m=0}^{\infty} \frac{\rho^m \log^n(a+m)}{a+m}, \quad \mathcal{D}_{n,k}^\rho(a) = \partial_a^k \ell_n(\rho, a), \quad k \geq 1.$$

At $\rho = 1$ set $\mathcal{D}_{n,k}^1(a) = \gamma_n^{(k)}(a)$, where this derivative is with respect to a . Only the differentiated family extends this way: the series defining ℓ_n does not generally converge at $\rho = 1$.

Introduce the reciprocal-gamma Appell polynomials

$$\frac{e^{xt}}{\Gamma(1+t)} = \sum_{n=0}^{\infty} Q_n(x) \frac{t^n}{n!}, \quad Q'_n(x) = nQ_{n-1}(x). \quad (7.1)$$

The first three are $Q_0 = 1$, $Q_1 = x + \gamma$, and $Q_2 = (x + \gamma)^2 - \zeta(2)$.

Proposition 7.1 (Appell–Laplace representation). *For $n \geq 0$, $k \geq 1$, $a > 0$, and $0 \leq \rho \leq 1$,*

$$F_{n,k}^\rho(a) := (-1)^{n+k} \mathcal{D}_{n,k}^\rho(a) = \int_0^\infty \frac{x^k e^{-ax}}{1 - \rho e^{-x}} Q_n(\log x) dx. \quad (7.2)$$

The integral, and every fixed number of a -derivatives, converge locally uniformly for positive a , uniformly in $\rho \in [0, 1]$. In particular $F_{n,k+1}^\rho = -\partial_a F_{n,k}^\rho$.

Proof. For the Lerch series $\Phi(\rho, s, a) = \sum_{m \geq 0} \rho^m (a + m)^{-s}$, parameter differentiation gives $\partial_a^k \Phi(\rho, s, a) = (-1)^k (s)_k \Phi(\rho, s + k, a)$. At $\rho = 1$ the analogous Hurwitz identity follows first from the convergent series and then by continuation; the Laurent pole at $s = 1$ is independent of a . Put $s = 1 + t$ and use the Mellin integral to get

$$k! \prod_{j=1}^k (1 + t/j) \Phi(\rho, k + 1 + t, a) = \frac{1}{\Gamma(1 + t)} \int_0^\infty \frac{x^{k+t} e^{-ax}}{1 - \rho e^{-x}} dx.$$

Here $\Phi(1, s, a)$ means $\zeta(s, a)$ in its convergent half-plane. Extracting the coefficient of t^n proves the formula. Near zero, $(1 - \rho e^{-x})^{-1} \leq (1 - e^{-x})^{-1} = O(x^{-1})$, so the integrand is bounded by a constant times $x^{k-1}(1 + |\log x|^n)$. At infinity, a bounded away from zero gives exponential decay. These estimates justify the extraction, all stated a -derivatives, and continuity at $\rho = 1$. $\square$

Lemma 7.2 (Strict real-rootedness of the Appell polynomials). *For every $n \geq 1$, Q_n has n distinct real roots.*

Proof. The Weierstrass product [10]

$$\frac{1}{\Gamma(1 + t)} = e^{\gamma t} \prod_{m=1}^{\infty} (1 + t/m) e^{-t/m}$$

converges locally uniformly. At any finite stage, coefficient extraction in (7.1) amounts to real translations and operators $1 + \partial_x/m$ applied to x^n . These operators preserve real-rootedness: away from roots of p , new zeros of $p + cp'$ solve $p'/p = -1/c$, whose left side is strictly decreasing between consecutive poles. Every new zero is simple; an inherited multiple zero loses one in multiplicity. Coefficient convergence and closure of monic real-rooted polynomials give real-rootedness in the infinite-product limit.

To prove simplicity, first remove the first $n - 1$ linear factors and their exponential translation factors from the product. The Appell polynomial for the remaining tail is still real-rooted by the same argument. Restore the removed factors. Translations commute with the differential operators and preserve multiplicity, while the $n - 1$ operators each lower inherited multiplicities and create only simple new zeros. Thus every final root is simple. $\square$

Proposition 7.3 (Global upper bound and persistence). *For every $n \geq 0$, $k \geq 1$, and $\rho \in [0, 1]$, $\mathcal{D}_{n,k}^\rho$ has at most n positive zeros counted with multiplicity. Once it has n distinct positive zeros at an order k , it has exactly n simple positive zeros at every higher order. With the zeros numbered increasingly,*

$$a_{j,k} < a_{j,k+1} < a_{j+1,k} \quad (1 \leq j < n), \quad a_{n,k} < a_{n,k+1}.$$

Proof. We use the elementary variation bound for Laplace transforms. If an integrable signed kernel has N sign changes, multiplication by $x_0 - x$ at a sign-change location removes one sign change. On its transform $G(a)$ this operation acts as

$$(\partial_a + x_0)G(a) = e^{-x_0 a} \frac{d}{da} (e^{x_0 a} G(a)).$$

For $N = 0$ the transform has fixed strict sign. Induction, together with Rolle's theorem counted with multiplicity, proves the bound of N zeros for general N ; it also proves nonvanishing of the transform. The same induction applies to every finite collection of zeros, so an infinite zero set cannot evade the bound. In (7.2), the positive weight leaves exactly the n sign changes of $Q_n(\log x)$. The domination already proved justifies the polynomial moments used in the variation argument.

Suppose now that $F_{n,k}^\rho$ has n distinct zeros. They are simple by the upper bound. Rolle's theorem gives a derivative zero between successive zeros. There is also a derivative zero after the last one: the function has a constant nonzero sign there and tends to zero at infinity, so it cannot depart from zero and then return toward zero without a stationary point. The identity $F_{n,k+1}^\rho = -\partial_a F_{n,k}^\rho$ supplies at least n distinct zeros at the next order. The upper bound supplies exact completeness and simplicity, and excludes any extra zero or coincidence with an old simple zero. Iterate this argument. The limit at infinity follows by dominated convergence in the Laplace representation, using any fixed positive reference a . $\square$

These are the manuscript's existing positivity and variation arguments. The new theorems below use their global multiplicity bound to turn local perturbation information and finitely certified signs into complete zero classifications.

8 The singular elementary endpoint of the Lerch deformation

The large-derivative theorem in the zero-geometry chapter of the pinned manuscript gives exactly n simple positive zeros once k is sufficiently large, uniformly for $0 \leq \rho \leq 1$. The following result describes a different limit: n and k are fixed and the Lerch parameter tends to its elementary endpoint $\rho = 0$. A multiple elementary zero generally sends most of its branches off the real axis. Consequently neither saturation nor the one-index monotonicity theorem extends indiscriminately to higher indices.

Use the notation of Section 7 and introduce the normalization normalization

$$G_{n,k}(a, \rho) = \frac{(-1)^{n+k}}{k!} \mathcal{D}_{n,k}^\rho(a) = \sum_{m=0}^{\infty} \rho^m f_{n,k}(a+m), \quad f_{n,k}(u) = \frac{P_{n,k}(-\log u)}{u^{k+1}}, \quad (8.1)$$

where

$$P_{n,k}(x) = n! [t^n] e^{xt} \prod_{j=1}^k (1 + t/j) = \prod_{j=1}^k \left(1 + \frac{1}{j} \frac{d}{dx}\right) x^n. \quad (8.2)$$

The series in (8.1) is locally uniformly absolutely convergent for $a > 0$ and $0 \leq \rho \leq 1$, because $k \geq 1$. It also defines a jointly holomorphic function for $\Re a > 0$ and $|\rho| < 1$, with the principal logarithm. The derivation of this exact spectral representation is already in the manuscript; it is not a new identity claimed here.

Lemma 8.1 (Elementary roots and a common compact zero region). *Fix integers $n \geq 1$ and $k \geq 1$. If $n > k$, put $d = n - k$. Then $P_{n,k}$ has a zero of multiplicity exactly d at $x = 0$ and k distinct strictly negative zeros. If $n \leq k$, all n zeros are distinct and strictly negative. There exist $0 < \epsilon < A < \infty$, depending on n, k , such that every positive zero of $G_{n,k}(\cdot, \rho)$, for every $\rho \in [0, 1]$, lies in $[\epsilon, A]$.*

Proof. For a real-rooted polynomial p and $c > 0$, the zeros of $p + cp'$ away from inherited multiple zeros solve

$$\frac{p'(x)}{p(x)} = -\frac{1}{c}.$$

The left side is strictly decreasing between its poles. There is one new simple zero between consecutive distinct zeros of p and one to the left of the smallest zero. An old root of multiplicity $r \geq 2$ is inherited with multiplicity $r - 1$; old simple roots are not inherited. Starting from x^n therefore gives the asserted root pattern through the first $k < n$ operations. At $k = n - 1$ the zero at 0 is simple; the next operation makes every root strictly negative, and all subsequent operations preserve simplicity and strict negativity. The exact coefficient at the inherited zero is

$$P_{n,k}(x) = \frac{n!}{k!d!}x^d + O(x^{d+1}) \quad (n > k). \quad (8.3)$$

Choose A strictly larger than the largest positive zero of $f_{n,k}$. For every $u \geq A$, $-\log u$ lies to the left of all the roots of the monic polynomial $P_{n,k}$, so $f_{n,k}(u)$ has the fixed strict sign $(-1)^n$. Each summand in (8.1) has that sign when $a \geq A$, including the nonzero $m = 0$ term.

At the other endpoint, $f_{n,k}(a) \sim a^{-k-1} \log^n(1/a) \rightarrow +\infty$ as $a \downarrow 0$. For $0 < a \leq 1$, the tail has a uniform absolute bound

$$\sum_{m=1}^{\infty} \rho^m |f_{n,k}(a+m)| \leq C_{n,k} \sum_{m=1}^{\infty} \frac{1 + \log^n(m+1)}{m^{k+1}} < \infty.$$

Consequently $G_{n,k}(a, \rho) > 0$ for all sufficiently small positive a , uniformly for $\rho \in [0, 1]$. This proves the common compact zero region; in particular, no unobserved real zeros can escape to 0 or infinity in the parameter limits below. $\square$

Theorem 8.2 (Complete small- ρ classification). *Fix integers $n > k \geq 1$ and put $d = n - k$ and*

$$\Delta_{n,k} = \frac{(-1)^d k! d!}{2^{k+1} n!} P_{n,k}(-\log 2). \quad (8.4)$$

Then $\Delta_{n,k} \neq 0$. For some $\rho_{n,k} > 0$, and every $0 < \rho < \rho_{n,k}$, all positive zeros of $G_{n,k}(\cdot, \rho)$ are simple and their exact number is

$$N_{n,k}(\rho) = \begin{cases} k+1, & d \text{ odd}, \\ k+2, & d \text{ even and } \Delta_{n,k} < 0, \\ k, & d \text{ even and } \Delta_{n,k} > 0. \end{cases} \quad (8.5)$$

Here the number is both the number of distinct positive zeros and the number counted with multiplicity, because simplicity is part of the conclusion.

More precisely, the k zeros corresponding to the nonzero elementary polynomial roots extend as real analytic simple branches in ρ . All d complex zero branches issuing from $a = 1$ have convergent Puiseux expansions

$$a_\nu(\rho) = 1 + z_\nu \rho^{1/d} + O_{n,k}(\rho^{2/d}), \quad z_\nu^d = -\Delta_{n,k}, \quad 1 \leq \nu \leq d. \quad (8.6)$$

The error describes fixed n, k ; no uniformity as the two integer indices vary is asserted. If $n \leq k$, the elementary endpoint already has n simple zeros and exactly n simple positive zeros persist for all sufficiently small nonnegative ρ .

Proof. The polynomial $P_{n,k}$ is nonzero and has rational coefficients. The classical Hermite–Lindemann theorem implies that $\log 2$ is transcendental: a nonzero algebraic logarithm would have transcendental exponential, whereas $\exp(\log 2) = 2$. Hence $P_{n,k}(-\log 2) \neq 0$. Only this nonvanishing consequence of classical transcendence theory is needed here; its source is Lindemann [11].

Put $\delta = a - 1$. Joint analyticity near $(a, \rho) = (1, 0)$ and (8.3) give

$$G_{n,k}(1 + \delta, \rho) = \delta^d A(\delta) + \rho B(\delta, \rho), \quad A(0) = \frac{(-1)^d n!}{k! d!}, \quad B(0, 0) = \frac{P_{n,k}(-\log 2)}{2^{k+1}}. \quad (8.7)$$

Set $\rho = s^d$. The function

$$H(z, s) = z^d A(sz) + B(sz, s^d)$$

is holomorphic near every point $(z, 0)$ with z bounded, and $H(z, 0) = A(0)(z^d + \Delta_{n,k})$ has exactly d distinct roots. The holomorphic implicit function theorem gives a convergent power series $z_\nu(s)$ through each of them. Thus $a_\nu = 1 + sz_\nu(s)$ proves (8.6). Choose a fixed small complex disk about $a = 1$ containing no other zero of $G_{n,k}(\cdot, 0)$. Uniform convergence on its boundary and Rouché's theorem show that it contains exactly d zeros, with multiplicity, for all sufficiently small ρ . The d branches just constructed account for all of them.

For real positive s , the branches through real z_ν remain real by uniqueness and complex conjugation, and the branches through nonreal z_ν remain nonreal by continuity. Every branch is simple, since $H_z(z_\nu(s), s) \neq 0$ for sufficiently small s . An odd-degree equation $z^d = -\Delta_{n,k}$ has exactly one real root. An even-degree equation has two real roots if $\Delta_{n,k} < 0$ and none if $\Delta_{n,k} > 0$. These real branches have positive a for small s .

The other k elementary zeros are simple and positive, so ordinary implicit continuation supplies exactly one real simple zero in each of k disjoint small disks about them. On the complement of these disks and the disk about 1 in the compact real interval supplied by Lemma 8.1, the continuous function $G_{n,k}(\cdot, 0)$ is bounded away from zero. Locally uniform convergence excludes additional real zeros there. This proves (8.5) globally, not merely near the elementary zeros. If $n \leq k$, the same argument with only simple-root disks proves the last assertion. $\square$

The sign of $\Delta_{n,k}$ is effective for any specified (n, k) : interval Horner evaluation of a rational polynomial at rational enclosures for $\log 2$ eventually separates it from zero. The nonvanishing theorem proves termination. The theorem does not give an explicit numerical value of $\rho_{n,k}$; such a value would require quantitative versions of the compact and local perturbation estimates.

Corollary 8.3 (First derivative: one or two zeros near the elementary endpoint). *For every fixed $n \geq 2$, $\mathcal{D}_{n,1}^\rho$ has exactly one simple positive zero for all sufficiently small $\rho > 0$ if n is odd, and exactly two if n is even. One zero is a real analytic perturbation of e^n . If n is even, the second is*

$$a_{\text{small}}(\rho) = 1 - \log 2 \left(\frac{n - \log 2}{4n} \right)^{1/(n-1)} \rho^{1/(n-1)} + O_n(\rho^{2/(n-1)}). \quad (8.8)$$

Proof. Here $P_{n,1}(x) = x^{n-1}(x + n)$, so

$$\Delta_{n,1} = \frac{(\log 2)^{n-1}(n - \log 2)}{4n} > 0.$$

Apply Theorem 8.2, noting that the unique nonzero elementary root is $x = -n$ and hence $a = e^n$. $\square$

Corollary 8.4 (Interior degeneracy is forced at indices three and four). *For each $n \in \{3, 4\}$ there exist $0 < \rho_* < 1$ and $a_* > 0$ such that*

$$\mathcal{D}_{n,1}^{\rho_*}(a_*) = 0, \quad \partial_a \mathcal{D}_{n,1}^{\rho_*}(a_*) = 0.$$

Proof. The manuscript's proved $\rho = 1$, $k = 1$ certificates give n simple positive zeros. They persist for $\rho < 1$ sufficiently close to 1: locally uniform convergence in both the function and its a -derivative gives a sign change and a nonzero derivative in a neighborhood of each old zero. The global bound of at most n zeros with multiplicity then gives exactly these n zeros. Corollary 8.3 gives respectively one and two zeros near $\rho = 0$. If every positive zero for $0 < \rho < 1$ were simple, the implicit function theorem and the common compact zero region would make the number locally constant throughout the connected interval $(0, 1)$. This contradicts the two endpoint counts. $\square$

This proves the existence of a multiple-zero event. It does not prove that an event is a unique nondegenerate saddle-node: assertions such as $\partial_a^2 G(a_*, \rho_*) \neq 0$ and $\partial_\rho G(a_*, \rho_*) \neq 0$ require additional analysis.

9 A rigorous obstruction to monotonicity of higher zero branches

Proposition 9.1 (Sensitivity of an elementary simple zero). *If $r > 0$ is a simple zero of $f_{n,k}$ and $a(\rho)$ is its analytic continuation as a zero of $G_{n,k}$, then*

$$a'(0) = -\frac{f_{n,k}(r+1)}{f'_{n,k}(r)}. \quad (9.1)$$

Proof. Differentiate $G_{n,k}(a(\rho), \rho) = 0$ at $\rho = 0$ in the normally convergent series (8.1). $\square$

Theorem 9.2 (The smallest zero can move to the right). *There exists $\rho_0 > 0$ such that $\mathcal{D}_{4,3}^\rho$ has exactly four simple positive zeros for $0 \leq \rho < \rho_0$. Its smallest zero $a_1(\rho)$ satisfies $a_1(0) = 1$ and*

$$a'_1(0) = \frac{P_{4,3}(-\log 2)}{64} > 0, \quad P_{4,3}(x) = x^4 + \frac{22}{3}x^3 + 12x^2 + 4x. \quad (9.2)$$

In particular, the statement that every positive zero decreases with ρ is false, even when all zeros under consideration are simple and the full allowed zero count is attained.

Proof. The polynomial has a simple zero at 0 and three distinct negative zeros by Lemma 8.1; hence the smallest elementary positive zero is 1. Its derivative in a is $f'_{4,3}(1) = -P'_{4,3}(0) = -4$ and $f_{4,3}(2) = P_{4,3}(-\log 2)/16$. Equation (9.1) gives the displayed value.

For an entirely rational sign proof put $L = \log 2$. The series

$$L = 2 \sum_{j=0}^{\infty} \frac{3^{-2j-1}}{2j+1}$$

gives $2/3 < L < 3/4$. Therefore

$$L^3 - \frac{22}{3}L^2 + 12L - 4 > \frac{8}{27} - \frac{22}{3} \frac{9}{16} + 8 - 4 = \frac{37}{216} > 0.$$

Since $P_{4,3}(-L) = L(L^3 - \frac{22}{3}L^2 + 12L - 4)$, its sign is strictly positive. Analytic continuation keeps the derivative positive on a sufficiently small interval. The root count and ordering follow from Theorem 8.2 with $n = 4, k = 3, d = 1$. $\square$

The accompanying rational interval calculation sharpens the coefficient to

$$0.0122109650329620 < a'_1(0) < 0.0122109650329621.$$

The rigorous sign does not depend on these decimal enclosures.

10 A full two-zero theorem and a nonmonotone branch

Lemma 10.1 (A single change of coefficient sign). *Let $\sum_{m \geq 0} |b_m| < \infty$. Suppose for an integer $M \geq 0$ that $b_m \leq 0$ for $0 \leq m \leq M$ and $b_m \geq 0$ for $m > M$. If $\sum_{m \geq 0} b_m < 0$, then*

$$\sum_{m \geq 0} b_m \rho^m < 0 \quad (0 < \rho \leq 1).$$

The same conclusion holds at $\rho = 0$ if $b_0 < 0$.

Proof. For $0 < \rho \leq 1$, divide the absolutely convergent series by ρ^M . For $m \leq M$, the inequality $\rho^{m-M} \geq 1$ and $b_m \leq 0$ give $b_m \rho^{m-M} \leq b_m$. For $m > M$, the inequalities reverse for the power but not for the final product: again $b_m \rho^{m-M} \leq b_m$. Summing proves

$$\rho^{-M} \sum_{m \geq 0} b_m \rho^m \leq \sum_{m \geq 0} b_m < 0.$$

At $\rho = 0$ the series equals b_0 . □

Theorem 10.2 (Two simple zeros throughout the deformation). *For every $\rho \in [0, 1]$, the first parameter derivative $\mathcal{D}_{2,1}^\rho$ has exactly two simple positive zeros $a_1(\rho) < a_2(\rho)$, and*

$$a_1(\rho) < \frac{13}{10} < a_2(\rho). \quad (10.1)$$

Both branches are continuous on $[0, 1]$ and real analytic on $(0, 1)$. The first branch is not monotone. More explicitly,

$$a_1(0) = 1, \quad a_1'(0) = -\frac{\log 2(2 - \log 2)}{8} < 0, \quad a_1(1) > 1. \quad (10.2)$$

Consequently this branch has a minimum at some point of $(0, 1)$.

Proof. The relevant elementary function is

$$f_{2,1}(u) = \frac{\log u(\log u - 2)}{u^2}.$$

At $a = 13/10$, put $b_m = f_{2,1}(13/10 + m)$. These coefficients are negative while $13/10 + m < e^2$ and positive while $13/10 + m > e^2$. There is therefore a finite integer M with the coefficient pattern in Lemma 10.1. Moreover $b_0 < 0$, since $1 < 13/10 < e^2$. There is no need to determine the numerical value of M .

The following exact rational enclosures come from the source's zero-certificate appendix [1]. The accompanying verification script reproduces them:

$$G_{2,1}(1, 1) \in [0.11418372, 0.11418373], \quad (10.3)$$

$$(13/10)^2 G_{2,1}(13/10, 1) \in [-0.12008911, -0.12008910]. \quad (10.4)$$

The positive normalization in the second line does not alter its sign. Lemma 10.1 now gives $G_{2,1}(13/10, \rho) < 0$ for every $\rho \in [0, 1]$. By Lemma 8.1 the sign is positive near $a = 0$ and for sufficiently large a , so there is a positive zero on each side of $13/10$. The manuscript's global Laplace variation bound is at most two zeros counted with multiplicity. Hence these are exactly the two zeros and both are simple.

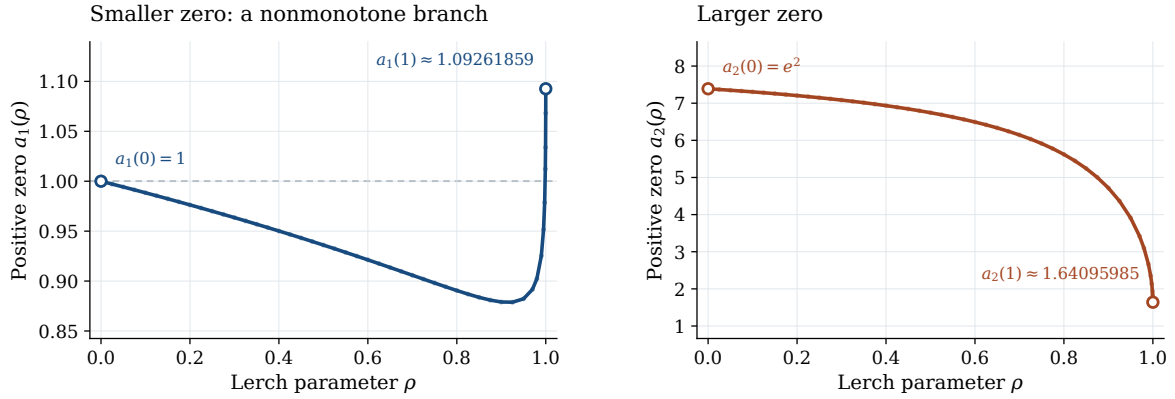
Simplicity gives analytic continuation locally in $\rho < 1$, and the fixed separating point $13/10$ preserves the branch ordering. Uniform convergence and the common compact zero region give continuity at $\rho = 1$. The analytic statement at 0 means that each branch has a real analytic extension to a small interval about 0. At $\rho = 0$ the smaller zero is 1; either (8.8) with $n = 2$ or direct implicit differentiation gives the derivative in (10.2). The signs (10.3) and (10.4) isolate a zero in $(1, 13/10)$. It must be $a_1(1)$ by the two-zero classification. Thus the branch first decreases below 1 and ultimately ends above 1. It is neither nondecreasing nor nonincreasing. Its continuity on the compact interval and these endpoint facts force its minimum into $(0, 1)$. □

Corollary 10.3 (The uniform saturation threshold at index two). *For every $k \geq 1$ and every $\rho \in [0, 1]$, $\mathcal{D}_{2,k}^\rho$ has exactly two simple positive zeros. For fixed ρ , they strictly interlace to the right as k increases. Thus the smallest possible uniform saturation threshold at index two is $K_2 = 1$.*

Proof. Theorem 10.2 supplies saturation at $k = 1$. Apply the manuscript's persistence and rightward-interlacing theorem, which follows from $F_{n,k+1}^\rho = -\partial_a F_{n,k}^\rho$ and the global bound with multiplicity. $\square$

The eleven rational endpoint signs replayed by the supporting script are exactly the two signs above and the nine signs supplying the source $n = 3, 4$ endpoint counts used in Corollary 8.4. The underlying outward integer Euler–Maclaurin routine is included verbatim, with its pinned repository provenance recorded in the output data.

Two zero branches of $\mathcal{D}_{2,1}^\rho(a)$



Numerical continuation at 51 parameters; selected residuals checked through a separate spectral formula. The exact two-zero theorem and nonmonotonicity proof do not depend on the plotted samples.

Figure 1: The two positive zeros of $\mathcal{D}_{2,1}^\rho(a)$, computed at 51 parameter values. The first branch is nonmonotone; the second joins e^2 to approximately 1.6409598465. The curves and interior locations are numerical diagnostics. The existence, simplicity, exact zero count, and nonmonotonicity of the first branch follow independently from the proofs.

The endpoint roots are numerically $a_1(1) \approx 1.0926185942922162563$ and $a_2(1) \approx 1.6409598465375973621$. Among the plotted samples, the smallest first-branch value occurs at $\rho = 0.925$, with $a_1 \approx 0.878826386000143658$. This is a sampled minimum, not a certified optimizer. The largest Laplace residual of the 102 roots is below 5.60×10^{-30} ; 14 independent Lerch/Hurwitz spectral checks have residuals below 3.13×10^{-30} . Numerical rounding is not interval-certified.

10.1 Monotonicity of the larger branch and a unique level crossing

The single change of coefficient sign also determines the shape of the larger zero branch and the part of the smaller branch above $a = 1$. These conclusions leave a sharper question about the number of extrema of the smaller branch below this level.

Corollary 10.4 (Shape constraints for the two zero branches). *Let $a_1(\rho) < a_2(\rho)$ be the two simple positive zeros in Theorem 10.2. Then a_2 is strictly decreasing on $[0, 1]$, with*

$$\frac{13}{10} < a_2(\rho) < e^2 \quad (0 < \rho \leq 1), \quad a_2(0) = e^2.$$

There is a unique $\rho_c \in (0, 1)$ such that $a_1(\rho_c) = 1$. Moreover,

$$a_1(\rho) < 1 \quad (0 < \rho < \rho_c), \quad a_1(\rho) > 1 \quad (\rho_c < \rho \leq 1),$$

and

$$a'_1(\rho) > 0 \quad (\rho_c \leq \rho < 1), \quad a'_2(\rho) < 0 \quad (0 < \rho < 1).$$

In particular, the level crossing is transverse, and every interior critical point of a_1 lies in $(0, \rho_c)$.

Proof. Write $G = G_{2,1}$ and

$$f(u) = \frac{\log u(\log u - 2)}{u^2}, \quad G(a, \rho) = \sum_{m=0}^{\infty} \rho^m f(a + m).$$

For a fixed $a \in [1, e^2]$ set $c = e^2 - a > 0$ and $b_m = f(a + m)$. The coefficient b_m is nonpositive for $m < c$ and nonnegative for $m > c$, and it vanishes if $m = c$. Thus

$$(m - c)b_m \geq 0 \quad (m \geq 0).$$

There are strictly positive terms, for example every integer $m > c$. Termwise differentiation is valid for $0 < \rho < 1$, and gives the strict differential inequality

$$\rho G_\rho(a, \rho) - cG(a, \rho) = \sum_{m=0}^{\infty} (m - c)b_m \rho^m > 0. \quad (10.5)$$

The use of a real cutoff c includes without exception the possibility that one coefficient vanishes at $a + m = e^2$.

For $\rho > 0$, the $m = 0$ term at $a = e^2$ vanishes and all terms with $m \geq 1$ are positive. Therefore $G(e^2, \rho) > 0$. Together with $G(13/10, \rho) < 0$ and the two-zero theorem, this places the larger zero in $(13/10, e^2)$, including at $\rho = 1$. The signs at the simple zeros are

$$G_a(a_1(\rho), \rho) < 0, \quad G_a(a_2(\rho), \rho) > 0,$$

since G is positive before the smaller zero, negative between the two zeros, and positive after the larger zero. At any zero in $[1, e^2]$, (10.5) gives $G_\rho > 0$. Implicit differentiation consequently gives $a'_2 = -G_\rho/G_a < 0$ for $0 < \rho < 1$. Continuity at both endpoints proves strict decrease on the whole closed interval.

Now fix $a = 1$ and put $c_0 = e^2 - 1$. The same inequality shows that

$$\frac{d}{d\rho}(\rho^{-c_0}G(1, \rho)) = \rho^{-c_0-1}(\rho G_\rho(1, \rho) - c_0 G(1, \rho)) > 0 \quad (0 < \rho < 1).$$

The expression is continuous up to $\rho = 1$. Since $f(1) = 0$ and $f(2) < 0$, one has $G(1, \rho) = \rho f(2) + O(\rho^2) < 0$ for small positive ρ . The exact certificate (10.3) gives $G(1, 1) > 0$. Strict increase of the weighted expression therefore gives precisely one zero $\rho_c \in (0, 1)$, with negative sign before and positive sign after it.

Because $1 < 13/10 < a_2(\rho)$, the sign of $G(1, \rho)$ is negative exactly when $a_1(\rho) < 1$, and positive exactly when $a_1(\rho) > 1$. This proves the asserted unique crossing and the inequalities on each side. When $\rho_c \leq \rho < 1$, the smaller root lies in $[1, 13/10)$, so (10.5) applies there as well. Its negative a derivative now gives $a'_1 = -G_\rho/G_a > 0$, including at the crossing itself. $\square$

The remaining shape question is whether a_1 has exactly one minimum in $(0, \rho_c)$; a sharper version asks whether its derivative has exactly one zero there. Neither uniqueness assertion is required for the corollary.

11 The logarithmic singularity at the Stieltjes endpoint

The opposite endpoint $\rho = 1$ is continuous for the differentiated Lerch family, but it is not infinitely differentiable in its deformation parameter. The following elementary Abel estimate identifies the first singular term and transfers it to every simple endpoint zero. Its logarithmic structure is the classical one for power-logarithm Dirichlet coefficients; the point here is its precise normalization and its consequence for zero branches.

Put $\tau = -\log \rho$ and $L = \log(1/\tau)$, so that $\rho \uparrow 1$ corresponds to $\tau \downarrow 0$. For $0 \leq j \leq k-1$ define the absolutely convergent moments

$$\mu_j(a) = \sum_{m=0}^{\infty} m^j f_{n,k}(a+m), \quad H(a, \tau) = \sum_{j=0}^{k-1} \frac{(-\tau)^j}{j!} \mu_j(a). \quad (11.1)$$

For $j = 0$ use $m^0 = 1$, including at $m = 0$.

Theorem 11.1 (The first singular Abel coefficient). *Fix $n \geq 1$, $k \geq 1$, and a compact interval $I \subset (0, \infty)$. Uniformly for $a \in I$, as $\tau \downarrow 0$,*

$$G_{n,k}(a, e^{-\tau}) = H(a, \tau) + \frac{(-1)^{n+k}}{k!(n+1)} \tau^k L^{n+1} + O_{n,k,I}(\tau^k L^n). \quad (11.2)$$

The family has continuous one-sided derivatives in τ through order $k-1$ at zero. Its next derivative satisfies

$$\partial_{\tau}^k G_{n,k}(a, e^{-\tau}) = \frac{(-1)^{n+k}}{n+1} L^{n+1} + O_{n,k,I}(L^n). \quad (11.3)$$

In particular it has no finite continuous k th derivative there. The leading singular coefficient is independent of a .

Proof. Monicity of $P_{n,k}$ and a uniform expansion at large m give

$$f_{n,k}(a+m) = \frac{(-1)^n (\log m)^n}{m^{k+1}} + O_{n,k,I} \left(\frac{(1 + \log m)^{n-1}}{m^{k+1}} \right), \quad m \geq 1. \quad (11.4)$$

Each a -derivative adds a factor $O(m^{-1})$, apart from bounded-degree logarithmic factors. All moments in (11.1) therefore converge normally on positive compact a -sets, including any fixed number of a -derivatives. Differentiation in τ up to order $k-1$ is justified by dominated convergence.

Write $R_{k-1}(x) = e^{-x} - \sum_{j=0}^{k-1} (-x)^j / j!$. For $0 \leq x \leq 1$,

$$R_{k-1}(x) = \frac{(-1)^k x^k}{k!} + O_k(x^{k+1}),$$

and for $x \geq 1$, $|R_{k-1}(x)| \leq C_k x^{k-1}$. Subtract H from the defining series for G and split the sum at $m = 1/\tau$. The leading part of the lower range is

$$\frac{(-1)^{n+k} \tau^k}{k!} \sum_{1 \leq m \leq 1/\tau} \frac{(\log m)^n}{m} = \frac{(-1)^{n+k} \tau^k}{k!(n+1)} L^{n+1} + O_{n,k}(\tau^k).$$

The Taylor remainder in that range is $O(\tau^{k+1} \sum_{m \leq 1/\tau} (1 + \log m)^n) = O(\tau^k L^n)$. The lower-order coefficient term in (11.4) contributes $O(\tau^k \sum_{m \leq 1/\tau} (1 + \log m)^{n-1} / m) = O(\tau^k L^n)$. Finally the upper range is bounded by

$$C \tau^{k-1} \sum_{m > 1/\tau} \frac{(1 + \log m)^n}{m^2} = O(\tau^k L^n).$$

These bounds prove (11.2), uniformly in a .

To avoid differentiating a remainder, compute the next derivative directly for $\tau > 0$:

$$\partial_\tau^k G_{n,k}(a, e^{-\tau}) = (-1)^k \sum_{m \geq 1} m^k e^{-m\tau} f_{n,k}(a+m).$$

Equation (11.4), the same cutoff, and $|e^{-x} - 1| \leq x$ for $0 \leq x \leq 1$ give

$$\sum_{m \geq 1} \frac{e^{-m\tau} (\log m)^n}{m} = \frac{L^{n+1}}{n+1} + O_n(L^n).$$

The lower logarithmic powers contribute $O(L^n)$. This proves (11.3) independently of differentiation of (11.2). $\square$

Theorem 11.2 (Endpoint regularity and singular displacement of simple zeros). *Let $r > 0$ be a simple zero of $G_{n,k}(\cdot, 1)$, and write $q = \partial_a G_{n,k}(r, 1) \neq 0$. There is a unique nearby real zero $a(\tau)$ of $G_{n,k}(\cdot, e^{-\tau})$ for all sufficiently small $\tau \geq 0$. Let $\alpha(\tau)$ be the real analytic zero of $H(\cdot, \tau)$ through $\alpha(0) = r$. Then*

$$a(\tau) = \alpha(\tau) - \frac{(-1)^{n+k}}{k!(n+1)q} \tau^k L^{n+1} + O_{n,k,r}(\tau^k L^n). \quad (11.5)$$

The branch is C^{k-1} , but not C^k , at the endpoint, in either τ or ρ . Moreover

$$\frac{d^k a}{d\tau^k} = -\frac{(-1)^{n+k}}{(n+1)q} L^{n+1} + O_{n,k,r}(L^n). \quad (11.6)$$

Proof. Uniform convergence of G and G_a , together with $q \neq 0$, gives one and only one zero near r , continuously in $\tau \geq 0$. The ordinary analytic implicit function theorem applies to the finite polynomial H , since $H_a(r, 0) = q$. Evaluating (11.2) at $\alpha(\tau)$ and using a uniform lower bound for $|G_a|$ first gives $a - \alpha = O(\tau^k L^{n+1})$. Taylor expansion in a , followed by $H_a(\alpha(\tau), \tau) = q + O(\tau)$, gives (11.5); the quadratic error in the displacement and the $O(\tau)$ change of the denominator are both $O(\tau^k L^n)$.

The convergent differentiated series show that G has all mixed derivatives of total order at most $k-1$ continuously at the endpoint. Implicit differentiation, or the one-sided implicit function theorem, therefore gives $a \in C^{k-1}$. Differentiate the implicit identity k times for $\tau > 0$. Every mixed derivative involving at least one a -derivative remains bounded, because that derivative improves the spectral tail by one inverse power of m . All derivatives of a of order below k have finite limits. Thus the only unbounded term other than $G_a a^{(k)}$ is the pure τ derivative in (11.3). Solving for $a^{(k)}$ proves (11.6). In this step replacing $G_a(a(\tau), e^{-\tau})$ by q incurs only $O(L^n)$: its difference is $O(\tau + |a(\tau) - r|)$, because $G_{a\tau}$ is bounded at the endpoint, and $|a(\tau) - r| = O(\tau + \tau^k L^{n+1})$. The leading coefficient is nonzero, so the k th derivative has no finite endpoint limit. Finally $\tau = -\log \rho$ is a smooth local change of variable with nonzero derivative at $\rho = 1$. $\square$

For the two branches of Theorem 10.2, this gives a particularly visible endpoint effect. If $r_i = a_i(1)$ and $q_i = \partial_a G_{2,1}(r_i, 1)$, then $q_1 < 0 < q_2$ and

$$a_i(\rho) = r_i + \frac{\tau}{3q_i} L^3 + O(\tau L^2), \quad \frac{da_i}{d\rho} = -\frac{1}{3q_i} L^3 + O(L^2).$$

Thus $a'_1(\rho) \rightarrow +\infty$ and $a'_2(\rho) \rightarrow -\infty$ as $\rho \uparrow 1$. The nonmonotonicity of a_1 already follows from the elementary endpoint and the rational Stieltjes signs; this Abel calculation supplies its quantitative endpoint behavior.

12 A complete exponential expansion for the angular zero

For integers $a \geq 1$ and real $b > 0$, write

$$H_j^{(b)} = \sum_{m=1}^j m^{-b}, \quad F_{a,b}(z) = \text{Li}_{a,b}(z, 1) = \sum_{n=2}^{\infty} \frac{H_{n-1}^{(b)}}{n^a} z^n.$$

The signed-density theorem in the pinned manuscript [1, `signed:thm:unique-zero`] proves that $\text{Im } F_{a,b}(e^{i\theta})$ has exactly one zero $\theta_{a,b}$ in the open upper semicircle, that it is simple, and that $0 < \theta_{a,b} < \pi/2$. Its proof uses a zero-mass density with one sign change and a strictly ordered Cauchy kernel. The source also proves a four-term expansion as $a \rightarrow \infty$, uniformly for $b \geq 1$, and explicitly leaves a systematic further expansion open.

We give every coefficient, with a remainder at any prescribed exponential scale. The result is uniform throughout $b > 0$. Only the elementary bound

$$1 \leq H_j^{(b)} \leq j \quad (j \geq 1, b > 0) \quad (12.1)$$

is needed for this stronger uniformity. The expansion is complete in the asymptotic sense. Theorem 12.4 proves that its infinite untruncated sum diverges at every fixed a, b , even after equal bases are combined.

12.1 The locally finite scale and its coefficients

Let $\mathcal{M}$ be the set of nonzero finitely supported sequences $\mathbf{m} = (m_3, m_4, \dots)$ of nonnegative integers. Put

$$|\mathbf{m}| = \sum_{n \geq 3} m_n, \quad q(\mathbf{m}) = \prod_{n \geq 3} (2/n)^{m_n}.$$

For every $0 < \eta < 1$, only finitely many $\mathbf{m}$ have $q(\mathbf{m}) > \eta$. Indeed, each occupied index satisfies $n < 2/\eta$, and $(2/3)^{|\mathbf{m}|} \geq q(\mathbf{m}) > \eta$ bounds the total multiplicity. Distinct multisets can give the same rational base: those coefficients must be added, not interpreted as distinct asymptotic scales.

Define the analytic germ

$$\phi_n(u) = -\frac{u \sin(n\pi/2 - nu)}{\sin(2u)} \quad (n \geq 3), \quad (12.2)$$

with its removable value at $u = 0$. Its Taylor coefficients are rational. For $k = |\mathbf{m}|$ define

$$C_{\mathbf{m}}(b) = \frac{(k-1)!}{\prod_{n \geq 3} m_n!} [u^{k-1}] \prod_{n \geq 3} \phi_n(u)^{m_n} \prod_{n \geq 3} (H_{n-1}^{(b)})^{m_n}. \quad (12.3)$$

Thus $C_{\mathbf{m}}(b)$ is an explicitly given rational polynomial in finitely many generalized harmonic numbers. In particular it is rational when b is a positive integer.

Theorem 12.1 (All exponential orders, uniformly in the second index). *For every fixed $0 < \eta < 1$, as the integer $a \rightarrow \infty$,*

$$\delta_{a,b} := \frac{\pi}{2} - \theta_{a,b} = \sum_{\substack{\mathbf{m} \in \mathcal{M} \\ q(\mathbf{m}) > \eta}} C_{\mathbf{m}}(b) q(\mathbf{m})^a + O_{\eta}(\eta^a), \quad (12.4)$$

uniformly for all real $b > 0$. The constant and the threshold for a depend on η , but not on b . The coefficient vanishes whenever the total multiplicity $\sum_{n \text{ odd}} m_n$ is even.

Proof. For $a \geq 4$, the Fourier series and its first angular derivative converge absolutely, uniformly under (12.1). The equation for the zero, normalized by 2^a , is

$$S_{a,b}(\delta) = \sin(2\delta) + \sum_{n \geq 3} H_{n-1}^{(b)}(2/n)^a \sin(n\pi/2 - n\delta) = 0. \quad (12.5)$$

The $n = 3$ term at $\delta = 0$ is $-H_2^{(b)}(2/3)^a$; the remaining nonzero terms there have size $O((2/5)^a)$ uniformly in $b > 0$. On $0 \leq \delta \leq 2(2/3)^a$, differentiating the series gives

$$S'_{a,b}(\delta) = 2 + O((2/3)^a).$$

For example, the differentiated tail is bounded by

$$\sum_{n \geq 3} n(n-1)(2/n)^a \leq (2/3)^a \sum_{n \geq 3} n^2(3/n)^4,$$

whose last sum is finite. The same elementary estimate bounds the undifferentiated tails. Since $1 \leq H_2^{(b)} \leq 2$, $S_{a,b}(0) < 0 < S_{a,b}(2(2/3)^a)$ for all sufficiently large a . Consequently the zero in this interval is unique and

$$\delta_{a,b} = O((2/3)^a), \quad S'_{a,b}(\delta) \geq 1 \quad (12.6)$$

there, uniformly in b . The source's global uniqueness identifies this local root with $\pi/2 - \theta_{a,b}$.

If $\eta \geq 2/3$, the sum in (12.4) is empty and (12.6) proves the assertion. Otherwise take $N = \lceil 2/\eta \rceil - 1 \geq 3$ and keep only the modes $3 \leq n \leq N$ in (12.5). On the real root interval the omitted tail is bounded by

$$\begin{aligned} \sum_{n > N} (n-1)(2/n)^a &\leq (2/(N+1))^a \sum_{n > N} (n-1)((N+1)/n)^4 \\ &= O_\eta(\eta^a). \end{aligned}$$

The derivative bounds give a unique small root δ_N of this finite equation and

$$\delta_{a,b} - \delta_N = O_\eta(\eta^a). \quad (12.7)$$

One can apply the mean value theorem between the two roots; both are in a common interval where the derivative of the finite equation is bounded below by 1.

Now regard $z_3, \dots, z_N$ as independent complex variables. The finite equation

$$\sin(2\delta) + \sum_{n=3}^N z_n \sin(n\pi/2 - n\delta) = 0$$

has a holomorphic solution $\delta = \Psi_N(\mathbf{z})$ near $\mathbf{z} = 0$, by the implicit function theorem with derivative 2. Dividing by $\sin(2\delta)/\delta$ gives

$$\delta = \sum_{n=3}^N z_n \phi_n(\delta).$$

Apply the finite-variable Lagrange–Bürmann formula, or insert an auxiliary scalar multiplying all z_n and use the ordinary formula. The coefficient of $\prod z_n^{m_n}$ is precisely

$$\frac{(k-1)!}{\prod m_n!} [u^{k-1}] \prod \phi_n(u)^{m_n}.$$

This is a finite-variable use of classical Lagrange inversion; Dirichlet-series inversion has a wider literature, for example Kuznetsov [12], but no infinite-variable convergence theorem is being assumed here.

Choose K so that $(2/3)^{K+1} \leq \eta$. Truncate the convergent Taylor expansion of Ψ_N at total degree K , then substitute $z_n = H_{n-1}^{(b)}(2/n)^a$. Its Taylor remainder is $O_\eta((2/3)^{(K+1)a}) = O_\eta(\eta^a)$, uniformly by the finite bounds $H_{n-1}^{(b)} \leq N$. Every discarded monomial whose base is at most η also has size $O_\eta(\eta^a)$. Every multiset with base greater than η is supported on $\{3, \dots, N\}$ and has total degree at most K . Combining this observation with (12.7) proves (12.4).

Finally, ϕ_n is even in u for odd n and odd in u for even n . Its product can have a coefficient at u^{k-1} only if $\sum_{n \text{ even}} m_n \equiv k-1 \pmod{2}$, equivalently if $\sum_{n \text{ odd}} m_n$ is odd. This proves the vanishing assertion. $\square$

Remark 12.2 (Why the proof truncates the Fourier modes first). The original Fourier series does not converge absolutely in a complex neighborhood of a real angular argument: high frequencies grow on one side of that neighborhood. An unqualified infinite-dimensional analytic implicit-function argument would therefore leave a real gap. The finite cutoff followed by a real derivative estimate avoids that gap and proves exactly the asymptotic assertion stated above. Theorem 12.4 proves that the formal sum over all multisets, and even the sum grouped by equal bases, diverges.

12.2 Ten terms and the next signed remainder

Write $h_j = H_j^{(b)}$. Let $T_{10}(a, b)$ be the sum of $c(q)q^a$ over the ten rows in Table 1. The first four rows recover the source's theorem. The next six and the uniformity for all $b > 0$ are supplied here.

Table 1: The complete angular expansion above the scale $(2/13)^a$. Equal rational bases have already been combined.

q	$c(q)$ in $T_{10} = \sum c(q)q^a$
$2/3$	$h_2/2$
$2/5$	$-h_4/2$
$1/3$	h_2h_3
$8/27$	$-23h_2^3/48$
$2/7$	$h_6/2$
$2/9$	$-(3h_2h_5 + h_8)/2$
$1/5$	$-h_3h_4$
$2/11$	$h_{10}/2$
$8/45$	$39h_2^2h_4/16$
$1/6$	$2h_2(h_3^2 + h_7)$

Corollary 12.3 (A sharp signed remainder after ten terms). *Uniformly for $b > 0$,*

$$\delta_{a,b} = T_{10}(a, b) - \frac{h_{12}}{2}(2/13)^a + O((4/27)^a). \quad (12.8)$$

In particular,

$$\frac{\delta_{a,b} - T_{10}(a, b)}{(2/13)^a} = -\frac{h_{12}}{2} + O((26/27)^a).$$

For all sufficiently large integer a , with a threshold independent of $b > 0$, the ten-term approximation strictly overestimates $\delta_{a,b}$ and therefore gives a strict lower approximation to $\theta_{a,b}$.

Proof. Apply Theorem 12.1 at the cutoff $4/27$. The only additional nonzero base strictly between $4/27$ and $2/13$, including $2/13$ itself, is $2/13$ with coefficient $-h_{12}/2$. This finite assertion can be checked by the multiset bounds in the proof, or by the exact generator supplied with the article. The source four-term coefficient and all six new displayed coefficients are also verified by substitution into the original Fourier equation, independently of the Lagrange coefficient formula. Finally $h_{12} \geq 1$, so the uniform remainder gives the eventual strict sign. $\square$

For illustration, at $a = 256$, $b = 4$, direct Fourier root evaluation gives

$$\delta_{256,4} \approx 4.42523791578002118187450681399 \times 10^{-46}.$$

The ten-term absolute error is approximately $4.24289006845 \times 10^{-209}$. Its normalized error is -0.541353663458730256 , approaching the predicted limit

$$-H_{12}^{(4)}/2 = -0.541076555073967515 \dots$$

The convergence of this last ratio is deliberately slow: its proved next scale is $(26/27)^a$. These decimal values are independent high-precision diagnostics, not interval certificates or inputs to the proof. The supplied script bounds the omitted Fourier tail and labels its floating-point status explicitly.

The next coefficient can also be made explicit. Extraction through the cutoff $1/7$ gives

$$\delta_{a,b} = T_{10}(a,b) - \frac{h_{12}}{2}(2/13)^a - \frac{27}{8}h_2^3h_3(4/27)^a + O((1/7)^a),$$

uniformly for $b > 0$. The nonzero contribution at base $4/27$ comes from the multiset $(3, 3, 3, 4)$; the other representations vanish by parity. The accompanying audit extraction substitutes all twelve terms back into the Fourier equation with exactly zero surviving residual above the cutoff. This extra negative term explains the direction of approach of the normalized numerical error.

12.3 An exact and finite coefficient algorithm

To compute the expansion above a prescribed rational cutoff η , enumerate nondecreasing multisets of integers $n \geq 3$ while $\prod(2/n) > \eta$. The total multiplicity and the possible indices are bounded as in the theorem. Remove multisets containing an even number of odd indices. Compute the coefficient in (12.3) with exact polynomial arithmetic and combine equal rational bases. This algorithm terminates for every cutoff, uses no integer-relation search, and returns the complete coefficient set at that scale.

For the cutoff $2/13$, it examines 25 multisets and produces the ten nonzero combined bases in Table 1. The verification routine then substitutes the answer into (12.5) in a separately implemented finite algebra of rational exponential bases. Every surviving coefficient cancels exactly. Thus the artifact checks the coefficient identities, while the theorem proves the uniform analytic remainder and the applicability of those identities to the actual zero.

12.4 The complete exponential series diverges at every fixed index

The finite-cutoff theorem cannot be strengthened to ordinary convergence of the complete coefficient series. This obstruction persists after all equal rational exponential bases have been combined. For a base q , put

$$\hat{C}(q; b) = \sum_{\mathbf{m}: q(\mathbf{m})=q} C_{\mathbf{m}}(b).$$

Each such sum is finite by local finiteness of the scale.

Theorem 12.4 (Divergence after grouping equal bases). *For every fixed integer $a \geq 1$ and real $b > 0$, the ordinary series*

$$\sum_q \widehat{C}(q; b) q^a$$

does not converge. In fact its individual terms are unbounded along an infinite sequence of distinct rational bases tending to zero. The ungrouped multiset series also diverges. Thus the complete expansion in Theorem 12.1 is asymptotic and necessarily divergent at every fixed a, b .

Proof. Choose an integer $s \geq 1$ with $2s > a$, and put $K = 2s + 1$. For an odd prime $P \neq 3$, consider the multiset with $2s$ copies of 3 and one copy of P . Its base is

$$q_P = \frac{2^K}{3^{2s}P}. \quad (12.9)$$

We first prove that this base has no other multiset representation. If k integers $n_1, \dots, n_k \geq 3$ give the same base, then

$$\prod_{j=1}^k n_j = 2^{k-K} 3^{2s} P.$$

Since the left side is an integer and $3^{2s}P$ is odd, $k \geq K$. Let $\Omega(n)$ denote the number of prime factors of n , with multiplicity. Additivity gives

$$\sum_{j=1}^k \Omega(n_j) = (k - K) + 2s + 1 = k.$$

Each summand is at least one, so every n_j is prime. The lower bound $n_j \geq 3$ excludes the prime two; hence $k = K$, and unique factorization leaves precisely $2s$ copies of 3 and one copy of P . Therefore $\widehat{C}(q_P; b)$ is exactly the coefficient of this one multiset, with no possible cancellation from other representations.

Write $h_j = H_j^{(b)}$, let $\sigma_P = \sin(P\pi/2) \in \{-1, 1\}$, and introduce the even analytic germ

$$B(u) = \frac{u}{\sin(2u)}, \quad B(0) = \frac{1}{2}.$$

For odd P , the coefficient germs in (12.2) satisfy

$$\phi_3(u) = B(u) \cos(3u), \quad \phi_P(u) = -\sigma_P B(u) \cos(Pu).$$

The multinomial prefactor in (12.3) is $(K-1)!/((2s)!1!) = 1$. Consequently

$$\widehat{C}(q_P; b) = -\sigma_P h_2^{2s} h_{P-1} \mathcal{P}_s(P), \quad (12.10)$$

$$\begin{aligned} \mathcal{P}_s(P) &:= [u^{2s}] B(u)^{2s+1} \cos^{2s}(3u) \cos(Pu) \\ &= \frac{(-1)^s}{2^{2s+1}(2s)!} P^{2s} + O_s(P^{2s-2}). \end{aligned} \quad (12.11)$$

The last equality is a polynomial identity in P followed by its leading-term estimate. Its highest power comes only from the u^{2s} coefficient of $\cos(Pu)$ and the constant coefficient 2^{-2s-1} of the remaining factor. All other powers are even and at most $2s-2$.

Combining (12.9) and (12.11) gives, along odd primes,

$$|\widehat{C}(q_P; b) q_P^a| = \frac{2^{K(a-1)} h_2^{2s}}{3^{2sa} (2s)!} h_{P-1} P^{2s-a} (1 + O_s(P^{-2})).$$

Here $h_{P-1} \geq 1$. Since $2s - a > 0$, these terms tend to infinity as $P \rightarrow \infty$ through odd primes different from three. Such primes are infinite; no estimate for their distribution is needed. The bases q_P are distinct and tend to zero. The necessary condition that individual summands tend to zero therefore fails, both for the grouped series and for the ungrouped series, and indeed for every ordering of these ordinary series. $\square$

For example, the first two polynomials in (12.11) are

$$\mathcal{P}_1(P) = -\frac{P^2 + 14}{16}, \quad \mathcal{P}_2(P) = \frac{P^4}{768} + \frac{11P^2}{48} + \frac{81}{32}.$$

The supplied script `check_angular_divergence.py` computes these coefficient polynomials exactly through $s = 6$, checks their leading coefficients, and compares six small cases with direct symbolic trigonometric expansion. The general proof above, not these finite checks, establishes divergence.

There is no conflict with the complete asymptotic theorem. Its cutoff is fixed first and then a tends to infinity. The divergent subsequence above fixes a first and lets the retained frequency grow without bound. Summation procedures beyond ordinary series convergence are a separate question.

13 Audit findings and proposed manuscript corrections

The research does not require withdrawing the source’s global zero bound, its eventual uniform saturation theorem, its $n = 1$ monotonicity theorem, or its all-conductor symbol kernel theorem. Those results survive the present analysis. The following corrections and status updates are narrower and should be integrated with their supporting proofs.

13.1 A logical wording correction in the rank conjecture

In `chapters/05-signed-kernels.tex`, Conjecture `signed:conj:rank` states two absolute ranks and then describes their difference as an equivalent statement. The difference follows from the two absolute formulas, but it does not recover them without another premise. Replace “Equivalently” by “Consequently” and “The equivalence” by “This implication.” This is a logical wording correction, independent of whether the conjecture is true. Theorem 2.1 now proves both absolute ranks, so after its proof has been integrated the conjecture can be promoted to a theorem.

The attached patch makes only those wording changes and the duplicated phrase correction below. It deliberately leaves the conjecture environment intact: installing a new theorem label without its proof and dependencies would be a poor intermediate repository state.

13.2 Boundary regularization must remain explicit

The removed coordinate $X_1^{0,1}$ and the retained difference row are essential parts of the matrix definition. A computation that discards both boundary rows, or retains them separately as homogeneous rows, solves a different problem. The canonical defect cY^n in Lemma 2.2 identifies exactly what the valid regularization preserves. The matrix generator includes column order, row labels, and the weight-five matrix itself so future implementations can compare the actual presentation, not merely a rank value.

Similarly, the new universal splitting concerns only the specified single-product coefficient system. Adding distribution or higher-depth relations can change the residual kernel. The theorem should not be paraphrased as a numerical dimension conjecture for all cyclotomic multiple polylogarithms.

13.3 The angular expansion is no longer an open construction

The paragraph following the source’s angular numerical table describes a systematic expansion beyond four terms as a separate research problem. Theorem 12.1 resolves that problem at every fixed exponential accuracy, and extends the uniformity from $b \geq 1$ to $b > 0$. The correct replacement should also mention Theorem 12.4: the infinite grouped series has unbounded terms at fixed a, b . Ordinary convergence is therefore ruled out, not merely unproved.

An analytic implicit-function argument on the entire complex Fourier series would be invalid without additional work: its frequencies grow exponentially off the real axis. The finite-mode argument in this article is part of the proof, not an implementation detail that may be dropped from an abbreviated presentation.

13.4 Higher-index monotonicity needs counterexamples, not extrapolation

The source proves a decreasing branch at $n = 1$. It does not thereby prove decreasing branches at larger indices. Theorem 9.2 gives a strictly positive initial velocity for the smallest $n = 4, k = 3$ branch. Theorem 10.2 gives an entire $n = 2, k = 1$ branch that is nonmonotone on $[0, 1]$. These are useful additions to the discussion of higher-index motion; they do not contradict the correctly restricted $n = 1$ theorem.

The regularity statement must distinguish the two endpoints. At a simple elementary zero, analytic dependence on ρ extends through 0. At a simple Stieltjes endpoint zero, Theorem 11.2 gives exactly C^{k-1} regularity, with a divergent k th derivative. An unqualified claim of analyticity on a neighborhood of the closed parameter interval would be false.

In the proof of the source result `half:zero:first`, replace “The endpoint signs in the endpoint signs and the Laplace representation above” by “The endpoint signs and the Laplace representation above.” This is an editorial duplication only.

13.5 Formal Herglotz obstructions and analytic evaluations

The general companion classification must retain the condition modulo $2e$. For example, the pair $e = 8, o = 3$ passes the corresponding tests modulo o and e , but fails $o^2 \equiv 1 \pmod{2e}$. A criterion that simply reuses the two moduli from the source’s F -classification would give an incorrect answer for $J(3/8)$.

The term “no reduction” in this setting means no reduction of the specified invariant pre-Bloch element. The boundary argument is not a transcendence theorem for actual values of J . Conversely, the positive cases of the formal theorem ensure existence in the stated calculus; they need not provide a short explicit logarithmic formula until a constructive five-term reduction is supplied. The consecutive family and $3/5$ evaluation provide such explicit analytic formulas in useful cases.

The adjacent-argument functional identity used for these evaluations is an independently derived reformulation of the known Choie–Kumar functional equation [8, Theorem 3.2(1)]. It should be credited accordingly. We do not claim priority for a general functional equation already present in that paper. The inspected follow-up work on character-weighted Herglotz integrals supplies relevant future directions [9].

13.6 Status synchronization across incoming reports

The recent rigidity report already proves the S_4 identity by an explicit combination of standard rows and a Möbius-duality row [3]. Older descriptions of S_4 as an unresolved global identity should be updated when that report is integrated. The earlier obstruction for a smaller relation

space remains a valid statement about that smaller space. It should be preserved with its scope, not treated as a contradiction to the later proof.

Likewise, the residual level-three and level-four invariant spaces were computed in the preceding report [2]. The new contribution is the free same-point lift and the full ranks under the manuscript's exact boundary convention. The supplied integration guide records this dependency to avoid duplicate priority claims.

14 A proposed research program

The results suggest concrete next problems. They are ordered by their direct dependence on the new structure, rather than by a claim that every direction has equal difficulty. Statements phrased as questions below are not theorems of this article.

14.1 Cyclotomic coefficient systems

1. Determine the residual kernel at general level. Theorem 3.1 removes every same-point sector canonically. What is the remaining Hilbert series of $\ker A_{N,w}^{-g}$ for fixed N ? At levels three and four, linear substitutions reduce the problem to small reflection groups. At levels five, six, and eight, the next step is to write the remaining color-polynomial equations, decompose them under color automorphisms, and look for a finite invariant-theoretic presentation. Exact matrices are valuable for selecting a candidate Hilbert series, but a proof must identify generators, relations, and all boundary defects.

2. Add distribution without losing the canonical boundary data. Which of the free sectors survive when distribution rows are added? The universal lift is supported on three colors per root pair, which makes the image of each distribution row explicitly computable. A useful first result would be a closed formula for the rank increment at prime-power levels. This would also clarify the relation between the coefficient systems here and the broader standard-relation spaces of [4].

3. Construct short relation certificates from the explicit dual basis. The nullspace test decides whether a proposed row is derivable, but does not by itself minimize the number or coefficient height of the rows used in a proof. Can the polynomial normal form be upgraded to a constructive reduction algorithm whose certificate length is polynomial in the weight? The existing S_4 certificate shows why the choice of extra argument transformations matters. A principled comparison between short certificates and large eliminated systems would be useful for future repository formalization.

14.2 Formal and analytic Herglotz arithmetic

4. Make every positive classification case constructive. The congruence test answers whether the invariant companion symbol vanishes, but a Bloch-group descent argument does not necessarily produce a compact five-term derivation. Find an explicit logarithmic formula, with a controllable number of terms, for all even-odd pairs obeying the two congruences. The families $J(e/(e^2 - 1))$ and $J(e/(e^2 + 1))$ are natural first targets beyond consecutive arguments. Their formal reductions are proved here; their simplest analytic evaluations remain to be found.

5. Classify subfield descent for combinations of symbols. An individual nonzero $\beta_M(a)$ cannot descend to a proper real subfield, but combinations can. Character decomposition of the Gram image should classify which coefficient vectors yield an element of $\Lambda^2 A(E)$ for a prescribed real subfield E . A complete answer would generalize the denominator-rank formula and could distinguish the smallest field needed by an explicit reduction from the ambient cyclotomic field.

6. Complete the rank problem for companion values at fixed denominator. Theorem 5.1 gives an exact denominator obstruction rank and a corresponding lower bound for formal companion values modulo rational dilogarithms. What additional rank is supplied by the numerator-conductor blocks? A useful formulation should fix a finite numerator set and a common cyclotomic field, then compute the intersection of the norm kernels. It should not confuse independence in a symbol space with numerical independence of the evaluated periods.

7. Replace the dyadic difference by other Hecke combinations. The proof uses a cyclic Laplacian for multiplication by 2 and a quadratic field norm between adjacent dyadic conductors. Analogous operators for an odd prime should interact with the Hecke functional equations of [7] and the character-weighted integrals of [9]. Identify a natural companion whose formal reduction is governed by a similarly explicit group operator, and determine its exact kernel. The first challenge is to keep rational wedge terms and conductor cancellation visible.

14.3 Lerch bifurcations and uniform zero counts

8. Prove or disprove uniqueness of the two-zero branch minimum. The smaller $n = 2, k = 1$ branch has at least one interior minimum and an endpoint slope tending to $+\infty$. Does it have exactly one critical point? Corollary 10.4 proves that the larger branch is strictly decreasing, and that the smaller branch increases after its unique crossing of $a = 1$. The remaining question therefore concerns only its motion below 1. Differentiating the implicit equation gives $a' = -G_\rho/G_a$; hence the first question reduces to the number of solutions of $G = G_\rho = 0$ on the first branch. A proof would require a new sign-variation statement for this pair of kernels. A visually single turn in the diagnostic plot is not enough to settle it.

Conjecture 14.1 (Unique stationary point of the smaller branch). For $n = 2, k = 1$, the derivative $a'_1(\rho)$ vanishes at exactly one point of $(0, 1)$.

The proved endpoint signs and Corollary 10.4 would then identify that point as the unique global minimum, below ρ_c . The numerical curve motivates the conjecture; no global exclusion of additional stationary points is asserted here.

9. Locate and classify the forced interior events at $n = 3, 4$. At $k = 1$, the small- ρ counts and the certified $\rho = 1$ counts differ by two. We prove that an interior multiple zero occurs. Determine whether there is exactly one event, and whether it is a nondegenerate saddle-node, by certifying

$$G = G_a = 0, \quad G_{aa} \neq 0, \quad G_\rho \neq 0.$$

Interval Newton or a two-variable argument principle could locate a candidate, but global exclusion of further events must be supplied separately. The current theorem deliberately makes no unverified nondegeneracy claim.

10. Determine the optimal uniform threshold K_n . The present result gives $K_2 = 1$. The elementary endpoint forces $K_n \geq n - 1$: when $k < n - 1$, the zero at $a = 1$ is multiple and uniform saturation by n simple zeros fails at $\rho = 0$. The natural sharp question is whether $K_n = n - 1$ for every $n \geq 2$. This equality is a candidate to test, not a conclusion of the existing large- k theorem. Exact sign certificates at selected separating points, extended uniformly in ρ by coefficient-sign arguments, could prove the first new cases. A counterexample would reveal where an interior loss of simplicity first occurs after the elementary endpoint has already split completely.

11. Make the small-parameter classification effective in both indices. The sign of $\Delta_{n,k}$ is decidable by exact rational intervals for $\log 2$, but the proof gives no uniform lower bound for its magnitude and no numerical value of $\rho_{n,k}$. Can one estimate the distances of the roots of $P_{n,k}$ from $-\log 2$ when n, k grow? Such an estimate, together with a quantitative Rouché argument, would turn the local classification into an explicit parameter range. The resulting sign diagram may have a useful asymptotic description when k/n tends to a fixed ratio.

12. Compute the full Abel expansion of the zero branches. The leading singular coefficient is independent of a , while later coefficients involve the tail expansion of $f_{n,k}(a + m)$ and the regular moments. Derive the complete expansion in powers of τ and $\log \tau$, including collision terms from implicit inversion. A particularly useful outcome would be a renormalized branch whose successive singular terms can be subtracted algorithmically, with an explicit differentiable remainder. This would also improve numerical continuation near $\rho = 1$.

14.4 Divergent angular expansions

13. Determine optimal truncation and the smallest attainable error. Theorem 12.1 keeps the cutoff fixed as $a \rightarrow \infty$, whereas Theorem 12.4 rules out ordinary convergence at fixed a . The next problem is quantitative: how should the cutoff depend on a to minimize the remainder, and what is the resulting error scale? The proof must control the growth of the finite implicit-function constants as the number of retained modes increases. The prime-frequency obstruction supplies explicit large coefficients and should constrain any proposed optimal rule.

14. Find a justified summation procedure. Can one recover the actual angular zero from the divergent coefficient family by a natural summation method? Possible approaches include keeping a finite Fourier equation exact while expanding only its low modes, or reorganizing terms by a parameter introduced before the mode cutoff. Any summability theorem must specify the order of summation and explain why the unbounded grouped terms do not invalidate the proposed procedure. A formal appeal to analyticity of the complete Fourier series is insufficient.

15. Extend the angular theorem to other colors and depths. The current local root is governed by a dominant Fourier mode and a nonzero real derivative, while the source's signed density proves global uniqueness. Which depth-three or non-Gaussian color families have both properties? Separate these two tasks: a local expansion can be constructed before global uniqueness is known, but identifying it with the intended geometric zero requires an independent global argument. Multi-sign-change densities may produce several roots and coupled singular limits, which would connect this direction with the Lerch bifurcation analysis above.

A practical order of attack. The most immediate projects are the first residual levels in item 1, constructive logarithmic evaluations in item 4, certified interior events in item 9, and the first optimal-truncation estimates in item 13. Each has a precise finite or analytic starting point supplied by this article. The broader period-independence questions remain important, but should be pursued with the formal-versus-numerical distinction preserved at every stage.

15 Verification, artifacts, and integration procedure

The verification has two roles. Exact finite calculations check the implementation of the displayed coefficient formulas, relation systems, and rational signs. Independent numerical calculations test the analytic evaluations and illustrate the geometry. Neither kind of computation replaces the proofs for arbitrary indices or conductors.

15.1 Exact coefficient and sign checks

Gaussian ranks.

The program `code/full_gaussian_rank.py` constructs the raw shuffle and stuffle rows directly, including the boundary difference. It independently performs rational elimination in weights 2 through 11, checks the explicit integer nullvectors in selected weights through 63, and reproduces the source's 92×23 matrix at weight five. Its receipt contains column order, row labels, the entire weight-five matrix, and its kernel basis.

Universal lift.

The program `code/universal_gaussian_lift.py` generates the cyclotomic rows afresh and verifies that the proposed lift annihilates every row at levels 3, 4, 5, 6, 8, 10, 12 in selected weights. It also checks the rank differences by independent rational elimination in weights 2, 3, 4 at levels 3 through 6.

Herglotz symbol calculations.

The program `code/verify_j.py` performs 32,764 exact first- and second-difference zero tests at odd conductors through 401. Independent rational elimination checks the two spans, their common rank, and the rank formula through conductor 101. The program also recovers the source's complete $J(2/q)$ classification from the general criterion, and checks the arithmetic of the displayed infinite families. These are integer group-algebra calculations; the logarithmic Gram nonvanishing is supplied by the analytic theorem, not by a floating-point determinant.

Lerch polynomial identities and endpoint certificates.

The program `code/verify_lerch_boundary.py` compares two exact constructions of $P_{n,k}$ in 152 cases. It computes 120 rational sign intervals for $\Delta_{n,k}$ through $n = 16$ and certifies the positive $n = 4, k = 3$ initial velocity. Eleven endpoint signs are replayed using the source's outward-integer Euler–Maclaurin routine. Two signs prove the $n = 2$ separator and endpoint displacement; nine provide the source's $n = 3, 4$ endpoint counts used to force interior multiple zeros. The upstream routine is included verbatim with its original repository path and SHA-256 recorded in the provenance data.

Angular coefficients.

The program `code/derive_angular.py` enumerates all rational exponential bases above a requested cutoff, computes their Lagrange coefficients exactly, combines collisions, and substitutes the result into an independently coded Fourier expansion. The supplied cutoffs $2/13$, $4/27$, $1/7$, and $1/10$ give respectively 10, 11, 12, and 20 nonzero combined bases. Every surviving symbolic residual vanishes exactly. The ten-term extraction examines 25 multisets.

Divergence coefficients.

The program `code/check_angular_divergence.py` computes the prime-frequency coefficient

polynomials through $s = 6$, verifies their leading coefficients, and checks six cases against direct trigonometric series expansion. Unique factorization and the general leading-term argument establish divergence; the finite checks corroborate the normalization.

15.2 Independent numerical diagnostics

The Herglotz verification uses 20 independent integrations at 100 decimal digits. For a positive rational p/q it substitutes $t = s^q$, so the integrand is

$$qs^{q-1} \frac{\log(1 + s^p)}{1 + s^q}.$$

It is regular at the lower endpoint and does not reuse the formula being tested. The absolute acceptance tolerance is 10^{-85} ; the largest observed residual in the supplied run is below 7×10^{-100} . This is high-precision corroboration, not an interval certificate.

The angular diagnostic evaluates the real Fourier root in 15 cases, with $a = 16, 32, 64, 128, 256$ and $b = 1, 2, 4$. It uses exact quadrant identities rather than subtracting trigonometric arguments close to integer multiples of $\pi/2$. An analytic bound controls the omitted Fourier tail; numerical rounding is not interval-enclosed. The theorem's uniformity over every real $b > 0$ follows from its proof, not from this finite sample.

The two-zero figure is also numerical. Its data and generating script are included, with independent spectral residual checks. The complete zero count, nonmonotonicity, and endpoint regularity follow from the proved sign and variation arguments independently of the plotted points.

15.3 Rebuilding the contribution

The archive has a single root directory. Its main components are `article/`, `code/`, `data/`, `integration/`, and `provenance/`. The complete source is `article/main.tex`; a flattened source is also supplied as `article/standalone.tex`. Both use the figures in `article/figures/`. From the archive root:

```
python code/run_checks.py
make pdf
```

The runner records package versions, script hashes, return codes, and output hashes in `data/validation_receipt.json`. The Python requirements are listed separately. The PDF build requires a standard LaTeX installation with the packages declared in the preamble and `latexmk`; no shell escape or network access is needed. The optional figure reconstruction is documented in the README.

The PDF was compiled to a stable set of references and its rendered pages were visually inspected. Package-level checks cover archive integrity, presence of the stated artifacts, and checksums. These checks concern the delivered files; they do not turn the mathematical arguments into machine-checked proofs.

15.4 Recommended integration order

First apply the small editorial patch, which is independent of the new proofs. Next integrate the Gaussian rank section with the universal extension and its exact matrix generator. Integrate the Herglotz classification before its denominator-rank and analytic corollaries. For zero geometry, retain the existing global bound and insert the elementary-endpoint, two-zero,

and Abel-endpoint results in that order. Finally replace the angular expansion’s open-status paragraph with the complete expansion and divergence theorems together.

The bibliography additions and the incoming-report status updates should accompany those insertions. All new labels use the prefixes `fg:`, `jnew:`, `lerchboundary:`, `lerchshape:`, and `angular:`. The repository was not modified by this contribution; the patch and fragments are proposed integration artifacts against the pinned commit.

References

- [1] V. Reshetnikov and contributors, *ProveIt: Analysis / Polylogarithms*, manuscript and research reports, repository snapshot of 10 October 2026, commit [a2a4cf58c49c745058c40e4a6748d472a3420f18](#). Pinned manuscript snapshot on [GitHub](#). The accompanying source manifest records the inspected chapter blobs.
- [2] ProveIt incoming research report, `polylogarithms_research_2026-10-10.zip`, at the snapshot above. In particular, the Gaussian section proves the residual level-three and level-four polynomial descriptions used here. [Incoming reports at the pinned snapshot](#).
- [3] ProveIt incoming research report, `polylogarithms_rigidity_20261010.zip`, at the snapshot above. In particular, the Gaussian section gives the Möbius-duality certificate for S_4 . The archive’s Git blob is recorded in the incoming manifest.
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